A Project Report

on

Design and Development of a Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester

A project report submitted in fulfillment of the requirements for Degree of the Bachelor of Science in Electronics and Communication Engineering.

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DECLARATION

I hereby attest that the project entitled **" Design and Development of a Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester"** represents our original work. It has not been submitted for evaluation elsewhere, whether in full or in part, for any academic degree. Any materials utilized from external sources have been duly acknowledged.

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ABSTRACT

Everything will be more advanced, appealing, and user-friendly in the modern world. The primary parts of all electronic circuits, integrated circuits (ICs) and transistors, have a wide range of applications. However, occasionally the circuit fails because of malfunctioning ICs or other common parts. Debugging the circuit and determining whether the circuitry is the issue or if the IC or transistor or other components is dead is, in fact, a very time-consuming task. Therefore, in order to identify these kinds of issues, we plan to create a project that would verify whether or not the transistor or IC or other common components in question is operating correctly. Therefore, this project's goal is to create an inexpensive, computer-independent, and intuitive digital logic integrated circuit (IC) and transistor tester. Basic digital logic ICs and large number of transistor and other common components can be tested by the logic IC and transistor tester. Arduino IDE has been used to develop the user interface to transmit and receive the instruction from computer to the Arduino through Universal Serial Bus (USB) interface for PC mode. The final outcome will be shown on the LCD. The logic IC functional tester has been constructed successfully and operates error-free using truth table matching method and the transistor or other components testing method was operates using the short circuit detection method. We have displayed the results of testing several IC types and also the short or open in two terminals in the result analysis chapter. The device is designed to be cost-effective, user-friendly, and practical for a range of electronic applications. Future improvements could expand testing capabilities to include operational amplifiers, further transistor types, and measurements of capacitance and inductance, making it a comprehensive tool for students and professionals in electronics.

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List of Abbreviations

IC: Integrated Circuit
Op-Amp: Operational Amplifier
LCD: Liquid Crstal Display
LED: Light Emitting Diode
TFT: Thin Film Transistor
OLED: Organic Light-Emitting Diode
I2C: Inter Integrated Circuit
XOR: Exclusive OR Gate
XNOR: Exclusive Not OR Gate
MPU: Micro Processor Unit
MCU: Micro Controller Unit
DMM: Digital Multimeter
LCR: Inductor Capacitor Resistor
ZIF: Zero Insertion Force
ADC: Analog to Digital Converter
MOSFET: Metal-Oxide-Semiconductor Field-Effect Transistor
UART: Universal Asynchronous Receiver Transmitter
I/O: Input Output
IGBT: Insulated Gate Bipolar Transistor
JFET: Junction Field-Effect Transistor
TRIAC: Triode for Alternating Current
Tx: Transmitter
Rx: Receiver
GND: Ground
Vin: Input Voltage
SDA: Serial Data Line
SCL: Serial Clock Line
SRAM: Static Random Access Memory
RAM: Random Access Memory
USB: Universal Serial Bus
GPIO: General Purpose Input Output
DAC: Digital to Analog Converter
EEPROM: Electrically Erasable Programmable Read-Only Memory
USART: Universal Synchronous and Asynchronous Receiver-Transmitter

CHAPTER 1

INTRODUCTION

1.1 Overview

In electrical and electronics engineering, the performance and reliability of components, especially integrated circuits (ICs) and transistors, are fundamental to the success of modern electronic systems. Digital logic ICs are central to numerous applications, ranging from simple logic functions to intricate control systems in industrial and consumer electronics. Given the importance of these components, engineers, technicians, and students must be equipped with reliable tools for verifying their functionality. Traditional methods of testing ICs and transistors, however, can often be cumbersome, time-consuming, or expensive, resulting in a demand for efficient and affordable testing devices that simplify and expedite the process.

To address this need, we have developed the Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester. This project is specifically designed to test both 14-pin digital logic ICs and a variety of transistor types, making it an essential tool for diagnosing a broad range of components using short circuit detection. By leveraging an ATMEGA328 microcontroller, a 16x2 I2C LCD display, and a BC547 transistor for short-circuit detection, the device is programmed with truth tables for common logic gates, including AND, OR, NAND, NOR, XOR, and NOT. The microcontroller systematically applies logical inputs (0 and 1) to the IC gates, compares the resulting outputs against expected values from these truth tables, and displays real-time results on the LCD. If all gates within an IC are confirmed to function correctly, the system displays the IC model number, indicating it is operational. Conversely, if any gate fails, the display highlights the specific fault, facilitating efficient troubleshooting and diagnosis. [1]

Moreover, the tester includes a short-circuit detection feature that enables it to test various transistor types as well as ICs. The BC547 transistor and LED indicators are employed to signal component status: a red LED lights up to indicate a faulty or shorted component, while a green LED confirms proper functionality. This short circuit detection mechanism not only allows users to test for faults across digital logic ICs but also extends functionality to include transistors, thereby enhancing the device's versatility. This multi-functional approach makes the tester a valuable, cost-effective tool, especially suited for educational, laboratory, and field applications where affordability and ease of use are prioritized.

Ultimately, the Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester ensures that only functional ICs and transistors are incorporated into larger systems, reducing the risk of failures caused by defective components. This project offers a practical solution for students, technicians, and engineers in electronics. As technology advances future expansions may include operational amplifier testing, capacitance, and inductance measurement, further solidifying its role as a comprehensive tool for electronics testing and education.

1.2 Limitation of Existing Works

The field of electrical and electronic component testing faces notable challenges, especially in testing digital logic ICs and transistors. Existing tools are often limited in scope, designed to test only specific IC models or types and lacking the flexibility to accommodate various logic gates or component types like transistors. This restriction forces users to depend on multiple devices, which increases complexity, inefficiency, and costs.

Many current testers are also bulky and over-engineered, often optimized for bulk testing instead of focusing on everyday use with smaller, more common logic gates such as AND, OR, NAND, NOR, XOR, and NOT. This makes it difficult for technicians and engineers to isolate faulty gates in small-scale projects or budget-sensitive environments, such as educational institutions. Additionally, most testers support only fixed gate configurations and cannot adapt to different logic structures or transistor types, further limiting their versatility. The high cost of these tools further restricts access for schools, hobbyists, and small-scale users. [11]

Addressing these limitations is essential to enhancing diagnostic precision, functionality, and affordability, thereby making testing tools more accessible to engineers, educators, and students alike.

1.3 Motivation

The limitations of existing digital logic IC testers highlight the need for a more versatile, affordable, and user-friendly solution that can accommodate a wider range of components, including transistors. Many current testers are specialized, expensive, or limited in scope, making them inaccessible to students, hobbyists, and professionals working in resource-constrained settings. This project is motivated by the need to create a cost-effective, comprehensive tool that addresses these challenges.

This project aims to develop a device capable of efficiently testing both digital logic ICs and various transistor types, with an integrated short-circuit detection feature to identify faulty components. By providing a reliable, accurate method for testing logic gates in 14-pin ICs and assessing transistor functionality, this project empowers users—particularly those with limited resources—to diagnose and validate components quickly and effectively. The goal is to combine functionality, precision, and affordability into one versatile device, making essential testing tools accessible to a broader audience and promoting innovation in the field of electronics.

1.4 Problem Statement

Current solutions for testing digital logic ICs are often expensive, limited in functionality, or lack the necessary flexibility, making it difficult for educational institutions, small-scale users, and professionals to access reliable testing equipment. Many existing tools are focused on testing specific gate types or are designed for bulk testing, failing to offer a comprehensive, cost-effective solution for everyday use.

To address this gap, a Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester is required, combining versatility and affordability. Utilizing the ATMEGA328 microcontroller, a 16x2 I2C display, and a BC547 transistor for short-circuit detection, this project focuses on developing a tool for testing 14-pin digital logic ICs. By incorporating pre-programmed truth tables, the device provides a reliable, immediate method for diagnosing faults in ICs and identifying short circuits, ensuring users can easily test and validate components.

1.5 Objectives

The primary objective of this project is,

- To develop a Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester specifically designed for testing digital logic ICs and identifying faulty components like transistors, rectifier, diode etc.
- To offer an affordable and accurate solution for testing gates commonly found in 14-pin ICs. To address a critical gap in the market for versatile and cost-effective testing tools accessible to professionals, students, and hobbyists.
- To utilize an ATMEGA328 microcontroller and a simple I2C 16x2 LCD display to provide real-time feedback and diagnostic results.
- To implement LED indicators to signal component faults (red for faulty, green for good), simplifying the troubleshooting process.
- To design a user-friendly device that accommodates users of varying expertise, ensuring accessibility, affordability, and accuracy while enhancing digital logic IC testing in educational and professional settings

By combining affordability with advanced functionality, it meets the growing demand for efficient diagnostic solutions in both educational and professional environments.

1.6 Conclusion

In conclusion, this project aims to develop a Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester to address the existing limitations in digital logic IC testing. Utilizing an ATMEGA328 microcontroller, a 16x2 I2C display, and a BC547 transistor for short-circuit detection, the device is designed to be cost-effective and versatile. Its key feature is the ability to test ICs using embedded truth tables, accurately verifying the functionality of logic gates, with results displayed clearly on the LCD screen.

This focus on digital logic ICs and short-circuit detection makes the device an ideal solution for educational institutions and hobbyists seeking an affordable and reliable testing tool. By filling a critical gap in the market, this project aims to advance electronic testing technology, improving the learning and testing experience for students and professionals alike.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter provides a comprehensive literature review related to the Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector. It explores existing technologies, methodologies, and advancements in electronic component testing, particularly focusing on digital logic integrated circuits (ICs). The review examines various testing techniques, including traditional approaches and modern microcontroller-based solutions, highlighting their strengths and limitations.

Additionally, this chapter discusses relevant studies that have contributed to the development of testing devices, emphasizing the importance of accuracy, reliability, and user-friendliness in electronic testing equipment. By synthesizing findings from previous research and industry practices, this literature review sets the foundation for understanding the current landscape of digital IC testing and informs the design and functionality of the proposed system.

2.2 Background Study

This chapter presents a comprehensive background study on advancements in microcontroller-based integrated circuit testers. It explores various studies that highlight innovative approaches, methodologies, and technologies in electronic component testing, focusing on the effectiveness, affordability, and user-friendliness of these systems. By synthesizing key findings, this review sets the foundation for understanding the current landscape of digital IC testing and informs the development of the proposed system.

2.2.1 Electronic Component Tester

This study by Elayan Abu Gharbyeh, Omar M. Alfroukh, and Bashar B. Ghannam (2021) presents a component tester built using an Arduino Mega 2560, designed to identify malfunctioning electronic components in laboratories and shops. [2]

Limitations:

- The system’s reliance on a PC application restricts its portability, limiting its effectiveness for field testing.
- The setup's complexity, involving hardware and software integration, may deter less experienced users, particularly beginners.

Case Study:

- Successfully implemented in educational laboratories, demonstrating effectiveness in identifying faulty components.

Problem Solution:

- User-friendly interface through the PC application simplifies the testing process for retail and educational environments.

2.2.2 Component and IC Tester Using 89S52 Microcontroller

The paper by Miss M. A. Tarkunde and Professor A. A. Shinde (2019) utilizes the 89S52 microcontroller to design an IC tester that verifies IC operation at the gate level. [3]

Limitations:

- The tester’s focus on the 74 series ICs restricts its use with other IC families, limiting its overall applicability.

Case Study:

- The system effectively evaluated circuit normalcy by switching bias voltages across PN junctions, helping confirm IC operation and fault detection.

Problem Solution:

- Gate-level testing offers detailed insights into IC functionality, addressing the need for precise evaluation methods.

2.2.3 Design and Development of Arduino Based IC Tester

This study by Md. Tanvir Nur et al. (2021) introduces a low-cost digital logic IC tester emphasizing testing basic 74 series TTL logic gates. This system offers an affordable solution focused on testing fundamental logic gates, making it especially suitable for educational environments. [4]

Limitations:

- The tester's simple design limits its ability to handle more complex ICs and lacks advanced testing features, which restricts its functionality to basic logic gate testing.

Case Study:

- Successfully used in educational settings, the tester helps students understand digital logic circuits by offering hands-on experience with IC testing.

Problem Solution:

- LCD screen for displaying results enhances user interaction and understanding of test outcomes.

2.2.4 Microcontroller Based IC Tester

This study by Mirza Shoaib Ahmed et al. (2021) presents an IC tester capable of delivering results in under a second, focusing on quick diagnostics to improve efficiency in component testing. [5]

Limitations:

- The emphasis on speed may reduce the depth of analysis, potentially limiting the accuracy of diagnostic information during testing.

Case Study:

- Tested in various electronic repair shops, showcasing rapid performance in identifying faulty components.

Problem Solution:

- Emphasizing energy efficiency and compact design addresses space and power consumption issues.

2.2.5 Scalable and Low-Cost Digital IC Tester

This study by Liakot Ali et al. (2021) discusses a low-cost IC tester that offers scan-path facilities for effective testing of PCB connections. This design provides an affordable solution aimed at improving connection verification and troubleshooting, particularly in educational settings. [6]

Limitations:

- May lack advanced features found in higher-end testers, limiting professional use capabilities.

Case Study:

- Successfully implemented in educational institutions for hands-on learning about circuit connections and troubleshooting.

Problem Solution:

- Scan-path facilities enhance versatility in identifying PCB errors, addressing critical needs in educational and practical applications.

2.2.6 Microcontroller Based Design of Digital IC Tester with Multi-Testing and Loop Testing Functions

This study by Maribelle et al. (2021) showcases a microcontroller-based IC tester capable of testing dual logic gates using the STC12C5A32 microcontroller chip. [1]

Limitations:

- Complexity of dual logic gate testing may require extensive user training compared to simpler testers.

Case Study:

- Implemented in electronics labs, facilitating advanced learning experiences with dual logic gate configurations.

Problem Solution:

- Enhancing the microcontroller's effectiveness addresses the need for sophisticated tools while remaining accessible to learners.

2.3 Conclusion

In conclusion, the literature review highlights significant advancements and methodologies in the field of electronic component testing, particularly for digital logic integrated circuits. The analysis of existing technologies reveals a trend toward microcontroller-based solutions that offer enhanced accuracy and efficiency while remaining cost-effective.

By identifying gaps in current testing methods and emphasizing the need for user-friendly devices, this chapter underscores the relevance of developing a Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector. The insights gained from this review not only inform the design choices made in this project but also pave the way for future innovations in electronic testing technology. Ultimately, this literature review serves as a crucial reference point for understanding how the proposed system can contribute to improving testing processes in both educational and practical applications.

CHAPTER 3

THEORETICAL BACKGROUND

3.1 Overview

Recent advancements in testing electronic components have highlighted precision, flexibility, and cost-efficiency, particularly for integrated circuits (ICs). Traditionally, testing relied on expensive automated systems, limiting accessibility in educational and small-scale environments. However, there is a shift towards affordable, microcontroller-based designs.

Microcontrollers like the ATMEGA328 enable the creation of cost-effective testers for digital logic ICs that use embedded truth tables to verify functionality. Innovations also incorporate short-circuit detection with transistors and LED indicators to identify faulty components. This approach is especially beneficial in educational settings, where budget-friendly tools enhance practical learning. By combining digital logic IC testing with short-circuit detection, these systems provide accessible solutions for students, hobbyists, and professionals.

3.2 Testing Components

In the realm of electronics, various types of testers are utilized to evaluate the functionality and parameters of components. Each type of tester serves a specific purpose, addressing different aspects of component testing. Below is a brief overview of some commonly used testers:

3.2.1 Component Tester

A component tester, sometimes referred to as an electronic part analyzer or component analyzer, is a tool used to assess and look into the specifications, performance characteristics, and operation of discrete electronic parts. These devices are typically used in the fabrication,

Figure 3.1 Electronic Component Tester

maintenance, and electronic engineering industries to ensure the dependability and quality of components before they are integrated into circuits or systems.

Component analyzers assess various electronic components, such as microchips, operational amplifiers, resistors, and transistors. They offer tailored testing methods for individual components or parts of complex circuits. Some advanced models include failure analysis modes for quickly identifying design flaws, though even sophisticated analyzers can malfunction.

3.2.2 LCR Meter

A device that measures the inductance, capacitance, and resistance of a component, sensor, or other device whose operation depends on capacitance, inductance, or resistance is called an LCR meter (Inductance (l), Capacitance (C), and Resistance (R)). LCR meters come in a range of styles, from tabletop to handheld. Handheld digital multimeters (DMMs) with capacitance measurement use a DC method to detect capacitance by measuring the DUT's RC time constant, achieving an accuracy of ±1%. They are lightweight, battery-operated, and portable.

Figure 3.2 LCR Meter

In contrast, benchtop LCR meters offer greater functionality, including computer control, automated data collection, programmable frequencies, and improved accuracy of up to 0.01%. They often feature sweep capability and DC bias options, making them ideal for production testing of sensors and components, dielectric constant measurements, and AC calibration of inductance, capacitance, and resistance standards. Overall, benchtop models provide more precise measurements than handheld alternatives.

3.2.3 Integrated Circuit Tester

Typically, an IC tester is small, sturdy, and simple to operate. It's a mobile gadget. Many digital integrated circuits with 6, 8, 10, 14, 16, 18, and 20 pins can be automatically tested by it. doesn't need any more hardware to be plugged in or added; all necessary gear is already installed. Before an IC is utilized in any circuit, it is tested or checked for condition using an integrated circuit tester, or IC tester. Many ICs are functionally tested, and the results are displayed as PASS/GOOD or FAIL/BAD, respectively. [7]

Figure 3.3 Integrated Circuit Tester

Analog integrated circuit testers measure characteristics like gain and distortion in components such as voltage regulators and operational amplifiers. Digital testers focus on microcontrollers and logic gates, analyzing input signals and output responses for correct logic functioning.

Universal IC testers combine both analog and digital capabilities, offering parametric measurements, functional testing, and component identification. These testers are crucial for ensuring the reliability of electronic circuitry across various applications.

3.3 Working of a Tester

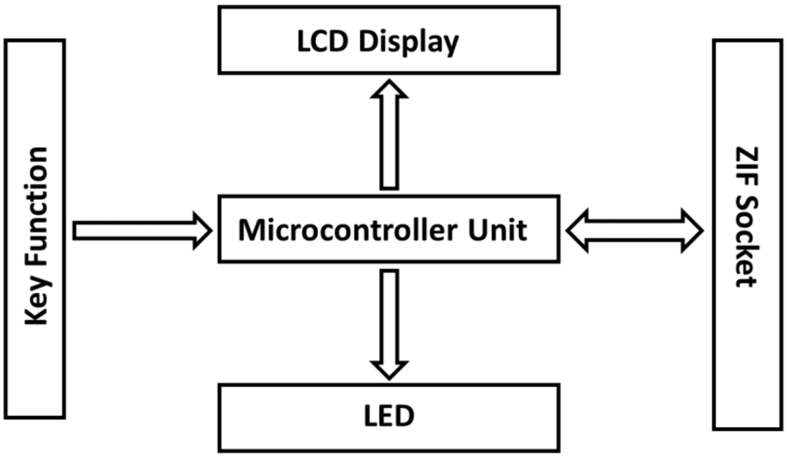


Figure 3.4 Basic Block Diagram of a Tester

A general electronic component tester evaluates various components, including integrated circuits (ICs), resistors, and capacitors. The tester features an LED indicator that signals when it is operational. Users insert the component into a 40-pin Zero Insertion Force (ZIF) socket and can reset the system to clear previous data before starting a new test. [8]

During testing, the microcontroller communicates with the Device Under Test (DUT) through the ZIF socket. It compares the DUT's responses to predefined outputs stored in its memory. If there’s a mismatch, an error message is displayed; if it matches, the LCD shows the component's identification number and "Good." The tester can detect common issues like open or shorted pins, providing insights into the component's health. Users can replace faulty components or reset the tester for another cycle. Overall, this versatile device simplifies the diagnostic process and enhances testing efficiency for a wide range of electronic parts.

3.4 Existing Tester

The literature review of comparable products completed by others is the main topic of this section. This market's expansion has led to a continuing trend of competition consolidation. It's critical to compare component testers available on the market to determine which features and gadgets people require. This will also allow component tester quality to be raised.

3.4.1 Kitek Universal IC Tester (DICT03)

Features: This tester evaluates over 1,500 ICs, including digital types like the 74 Series and 8085, as well as analog ICs such as op-amps and voltage regulators. It tests common cathode and anode 7-segment displays and features an auto-search function for digital ICs, a 40-pin DIP ZIF socket, a 50-key keypad, and a 16x2 LCD display.

Figure 3.5 Kitek Universal IC Tester (DICT03)

Limitations: Very expansive, very heavy, not portable, limited to IC family only.

3.4.2 MME-ECT20 Electronic Component Tester

Features: Testing equipment including IC Testers, Component Testers, Microprocessor trainer kits based on 8085, 8086, Z80, 8088 Microprocessors, Microcontroller trainer kits based on 8051/8052, 689HC11 Microcontrollers, VLSI/FPGA based educational systems, ARM-7 based development kits. This electronic component tester can test analog ICs up to 20 pins and components like diode, transistor etc.

Figure 3.6 MME-ECT20 Electronic Component Tester

Limitations: Very expensive, very heavy, not portable.

3.4.3 LCR-TC1 Multifunctional TFT Tester

Features: This tester detects NPN and PNP transistors, capacitors, resistors, diodes, MOSFETs, IGBTs, JFETs, TRIACs, power sources, infrared waveforms, and Zener diodes. It includes a self-calibration function.

Figure 3.7 LCR-TC1 Multifunctional TFT Tester

Limitations: Can’t provide value, lower accuracy, can only test electrical component.

3.5 Importance of Testing

Component testing is vital for ensuring the reliability, safety, and performance of electrical and electronic systems. By systematically evaluating components before they are integrated into larger systems, testing helps identify defects, verify operation, and ensure compliance with quality standards. [9] [11]

Reliability Assurance: Testing confirms that every component functions correctly before being integrated, reducing the risk of system failures. By catching potential issues early, engineers can ensure greater reliability in the final product.

Fault Detection: Early detection of defects through testing helps prevent costly repairs and ensures that components meet quality benchmarks. Addressing faults early enhances the overall integrity and dependability of the system.

3.6 Testing Methods

There are many different ways to test electrical and electronic components to make sure they are functional and reliable. [10] [11]

Continuity Testing: This method verifies that current flows without interruption through a circuit. It helps identify broken connections or open circuits.

Resistance Testing: Resistance testing measures a component's resistance to detect faults such as short or open circuits. It ensures components are functioning within expected resistance values.

Voltage Testing: Voltage testing identifies issues like voltage drops or short circuits by measuring voltage levels across components. It helps ensure components operate within their specified voltage ranges.

Current Testing: Current testing measures the flow of electrical current through components such as diodes and transistors. It assesses whether these components are functioning correctly and delivering the expected performance.

Frequency Response Testing: This testing evaluates how components respond to various frequencies. It provides insights into a component's bandwidth and signal handling capabilities.

3.7 Conclusion

The theoretical background chapter has provided key insights into various component tester models, enhancing our understanding of the challenges and advancements in electrical and electronic testing. By analyzing several research articles, we clarified effective technologies and methodologies to address these challenges, guiding our design choices for the proposed component tester tailored to professionals and enthusiasts.

This chapter establishes a framework for our research and development efforts, directing us toward creating a reliable and adaptable component tester that meets industry standards while improving user experience. By identifying gaps in current practices, we aim to contribute a versatile tool that enhances diagnostic capabilities and efficiency in testing electronic components, supporting the growing demand for effective solutions in a complex electronic landscape.

CHAPTER 4

BASIC HARDWARE AND CIRCUITRY

4.1 Overview

This chapter presents the design and development of the Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector. Central to the device are the ATMEGA328 microcontroller, a BC547 transistor circuit for short-circuit detection, and an I2C 16x2 LCD display, providing an efficient solution for testing digital logic integrated circuits (ICs) and identifying faulty components. The tester emphasizes simplicity and affordability, focusing on evaluating logic ICs using predefined truth tables while ensuring a user-friendly experience.

In addition to logic IC testing, the device features short-circuit detection with LED indicators: a green LED signals a good component, while a red LED indicates a fault. This combination of functionalities makes the tester accessible for both educational and practical applications, ensuring reliability without unnecessary complexity. This chapter will outline key hardware components, operational principles, and advantages of this targeted approach compared to more complex systems, as well as discuss potential future improvements to enhance its capabilities and usability.

4.2 General Block Diagram

The general block diagram of proposed system consists of MCU, Socket, Keypad, Display, External Circuitry and Power Supply Unit, this are the key components for the project.

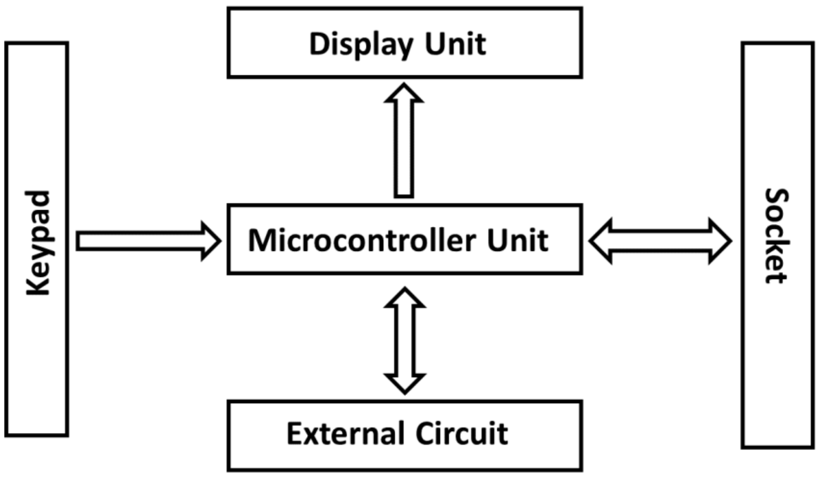


Figure 4.1 General Block Diagram of the Proposed Project.

Microcontroller Unit (MCU): Microcontroller is the central processing unit that executes testing algorithms and controls the operation of the device.

Socket: Socket allows for secure and easy insertion of ICs during testing.

Keypad: The keypad is an input device used for selecting testing modes.

Display Unit: Display unit shows test results and operational information to the user.

Power Supply Unit: This unit provides the necessary electrical power to all components.

External Circuitry: This portion is constructed using BC547 Transistor to test the continuity.

4.3 Microcontroller Unit (MCU)

The MCU acts as our tester's brain, sending signals to parts, receiving replies, and making judgments on component quality. Microcontrollers are common in modern electronics, from home appliances to industrial machinery, each with unique features and uses. Here are some popular microcontroller platforms:

Arduino: Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board and a piece of software, or IDE that runs on computer, used to write and upload computer code to the physical board.

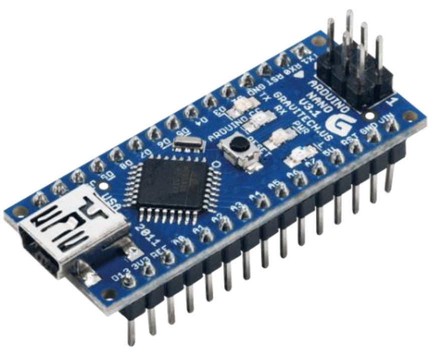


Figure 4.2 Arduino Nano

Arduino is a favorite for beginners in electronics, and for good reasons. Unlike other programmable boards, it doesn't require a separate programmer—just a USB cable for loading new code. The Arduino IDE uses a simpler version of C++, making programming easier to grasp. Plus, Arduino offers a standard layout that makes microcontroller functions more accessible.

ESP (Espressif Systems): Due to their integrated Bluetooth and Wi-Fi, ESP microcontrollers—especially those in the ESP8266 and ESP32 families—have become more and more well-liked. These microcontrollers are widely utilized in Internet of Things (IoT) projects, enabling wireless communication between devices and internet connections.

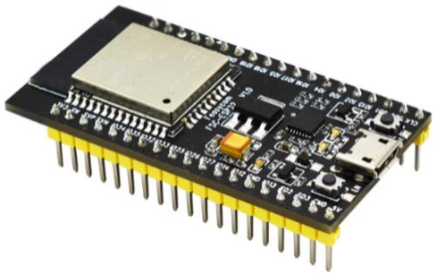


Figure 4.3 ESP32

A vibrant community supports ESP-based boards, which are feature-rich, reasonably priced, and well-supported. Numerous applications, including as remote monitoring systems, sensor networks, and home automation, are appropriate for them.

Raspberry Pi: Although not strictly speaking a microcontroller in the conventional sense, the Pi is a potent single-board computer that finds frequent use in embedded and Internet of Things applications. When it comes to connection, memory, and processing capability,

Figure 4.4 Raspberry Pi 3

Raspberry Pi boards outperform conventional microcontrollers. Their ability to run a complete operating system (often a variation of Linux) allows them to do more sophisticated functions including online browsing, playing multimedia, and managing server software. For applications requiring more processing power, such media centers, home servers, and do-it-yourself electronics projects, Raspberry Pi boards are well-liked.

4.3.1 Selecting the Right MCU

Comparing the advantages of Raspberry Pi, ESP boards, and Arduino is essential for project selection. Raspberry Pi is best for power-intensive tasks and running Linux, while ESP boards like the ESP8266 and ESP32 excel in Internet of Things (IoT) projects due to their built-in Bluetooth and Wi-Fi capabilities. Arduino stands out for its reliability, user-friendliness, and extensive community support, making it ideal for electronics projects.

Overall, Arduino is the simplest and most accessible option, particularly for beginners, thanks to its compatibility with various components and abundant online resources. While Raspberry

Pi and ESP boards have their strengths, Arduino remains the top choice for straightforward electronics applications.

Table 4.1 Arduino Mega vs Uno Vs Nano

Feature	Arduino Uno	Arduino Nano	Arduino Mega
Microcontroller	ATmega328P	ATmega328P	ATmega2560
Digital I/O Pins	14	14	54
Analog Pins	6	8	16
Flash Memory	32 KB	32 KB	256 KB
SRAM	2 KB	2 KB	8 KB
EEPROM	1 KB	1 KB	4 KB
Clock Speed	16 MHz	16 MHz	16 MHz
Price (approx.)	TK. 700-750	TK. 450-500	TK. 1700-1900

The Arduino Nano is the board that is most suited for this project out of the Arduino Uno, Nano, and Mega. The Nano has the same capabilities as its larger equivalents but being smaller in size. It has enough processing power, flash memory, and digital and analog ports for most projects.

4.4 Socket

A mechanical part used to make it easier to insert and remove integrated circuits from a printed circuit board (PCB) without soldering is called an IC (Integrated Circuit) socket, sometimes referred to as a chip socket or integrated circuit socket. To support varying integrated circuit package sizes and pin counts, IC sockets are available in a variety of types and layouts.

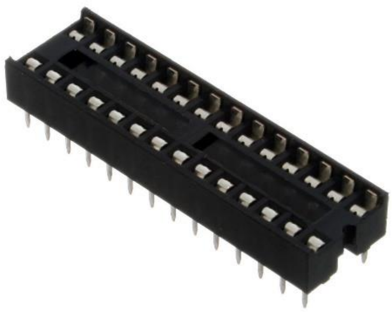


Figure 4.5 DIP Socket

Dual In-Line Package (DIP) Socket: ICs with dual in-line package designs, in which the pins extend from two opposite sides of the package, are intended to use these sockets. A lot of hobbyists and prototyping applications use DIP sockets.

Figure 4.6 ZIF Socket

Zero Insertion Force Socket (ZIF): ZIF sockets have a mechanism that lets you insert or remove the IC without using any force on the pins. By doing this, the possibility of breaking the IC or PCB during installation or replacement is reduced. ZIF sockets are frequently utilized in settings like test and development environments where it is necessary to remove and replace integrated circuits (ICs) on a regular basis.

4.5 Keypad

The keypad is a vital component of the tester, enabling intuitive user interaction and operation. Its design enhances usability, ensuring that users can easily navigate menus and initiate tests.

The push button interface offers a straightforward and reliable means for user engagement. These buttons are affordable and easy to use, requiring minimal components for installation. Their simplicity ensures that even users with limited technical knowledge can operate the device intuitively, making them an ideal choice for enhancing usability in various applications.

Figure 4.7 Membrane Keypad

The push button interface offers a straightforward and reliable means for user engagement. These buttons are affordable and easy to use, requiring minimal components for installation.

Their simplicity ensures that even users with limited technical knowledge can operate the device intuitively, making them an ideal choice for enhancing usability in various applications.

Overall, the push button keypad is more cost-effective and easier to set up compared to the membrane keypad, making it a practical choice for this project.

4.6 Display Unit

Microcontrollers are frequently used to interface with different kinds of display devices in order to communicate with users or provide visual feedback. The following are a few typical display unit types that pair with microcontrollers:

Figure 4.8 LCD Display

LCD (Liquid Crystal Display): One of the most often utilized display technologies in microcontroller-based devices is LCD (Liquid Crystal Display). Character LCDs, which are typically used to display alphanumeric characters, and graphical LCDs, which may display graphics and images, are among the sizes and configurations in which they are available. Character LCDs are frequently employed in applications including industrial control panels, consumer electronics, and embedded systems where text-based information is sufficient since they are relatively easy to interface with microcontrollers.

Figure 4.9 OLED Display

Organic Light-Emitting Diode (OLED) Display: OLED screens are gaining popularity because of its low power consumption, broad viewing angles, and excellent contrast. OLEDs provide superior contrast and deeper blacks than LCDs because they emit light directly, negating the need for a backlight. OLED screens are widely used in digital cameras, wearable technology, smartphones, and portable gadgets. They come in a variety of sizes and resolutions. They are ideal for applications needing vivid colors and high-resolution graphics.

The tester features a 16x2 I2C LCD display for real-time feedback on operational amplifiers, inductor and capacitor values, and digital logic ICs. Its clear interface and reduced wiring make it ideal for both professional and educational use.

4.7 Power Supply Unit

The power supply unit (PSU) of the Tester is essential for providing the necessary electricity to operate the device effectively. A reliable power source ensures that all components function properly, enabling accurate testing and diagnostics.

The tester typically utilizes a power adapter that converts AC mains voltage into the required DC voltage. This setup not only enhances portability but also simplifies usage by eliminating the need for complicated wiring or frequent battery replacements. Users can easily connect the tester to standard electrical outlets in various environments, ensuring consistent and reliable operation without the worry of running out of power during critical tasks. The convenience of a power adapter allows users to focus on their testing activities without interruptions.

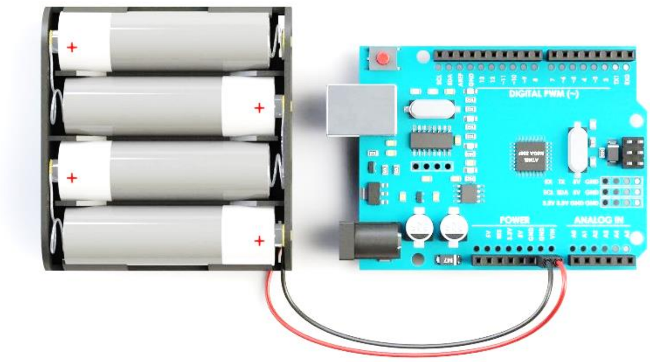


Figure 4.10 Battery Pack for Powering Arduino

A battery equipped with a Battery Management System (BMS) may be used in these circumstances. Increased portability and independence from external power sources are two benefits of this approach, although there are higher costs involved. By including a BMS, you can be sure that the battery will be properly managed and protected, protecting against potential hazards like overcharging and over discharging.

The decision between a battery with BMS and a power adapter ultimately comes down to the unique needs and preferences of the users, taking into account aspects like cost, portability, and convenience.

4.8 Printed Circuit Board (PCB)

A printed circuit board (PCB) is essential in electronic systems, providing a substrate for mounting and connecting components while ensuring efficient signal routing. A prototype board, or Veroboard, is a fundamental tool in electronics prototyping, featuring a grid of holes for connecting components, allowing rapid circuit iteration and debugging.

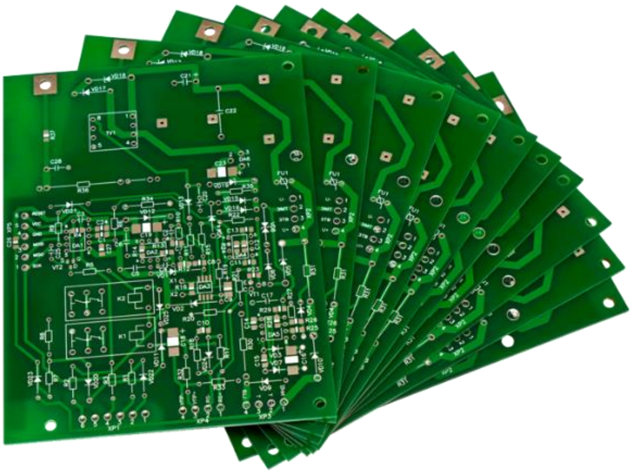


Figure 4.11 Custom Printed Circuit Board

Computer-aided design (CAD) software is crucial for designing custom PCBs tailored to specific project requirements. The layout includes component placement, trace routing, and features like power planes and signal layers. Once designed, the PCB is manufactured by etching the circuit layout onto a substrate and covering it with conductive copper. After manufacturing, electronic parts are soldered onto the board. Custom PCBs are ideal for long-term installations and mass production due to their scalability, and professional appearance.

4.9 Conclusion

In conclusion, this chapter has provided a thorough examination and comparison of various hardware components essential for the development of the Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector. By evaluating options such as microcontrollers, displays, keypads, and power supplies, we have identified the most suitable components based on criteria like cost, usability, compatibility, and reliability.

The selected components not only meet the technical requirements of the project but also enhance user interaction and overall functionality. The microcontroller offers sufficient processing power, while the user-friendly keypad and display facilitate intuitive operation for users of all skill levels. Additionally, the reliable power supply ensures consistent performance across all components. This careful selection process establishes a solid foundation for the project's subsequent phases, ensuring that each component works harmoniously within the system. Ultimately, the insights gained from this chapter will guide future development efforts and contribute to creating a robust and efficient testing solution for electrical and electronic components.

CHAPTER 5

OVERVIEW OF THE SYSTEM

5.1 Overview

This chapter analyzes the Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector, focusing on its design and capabilities in addressing challenges in testing integrated circuits (ICs). We outline the system's primary goals, emphasizing the specific problems it aims to solve in digital logic IC testing and component evaluation. Key components include the ATMEGA328 microcontroller, a 16x2 I2C LCD display, a secure socket for IC insertion, and external circuitry using a BC547 transistor for short-circuit detection. The chapter discusses how these elements work together for accurate testing and reliable output, along with user interactions like inserting ICs and interpreting results.

We also examine the technologies and development methods employed, including programming languages for microcontroller programming and logic analysis. The use of embedded truth tables and short-circuit detection mechanisms is explained, providing insight into the system's technical foundation. Overall, this chapter clarifies the system's goals and functionality, setting the stage for performance evaluation and potential improvements in subsequent chapters.

5.2 System Design

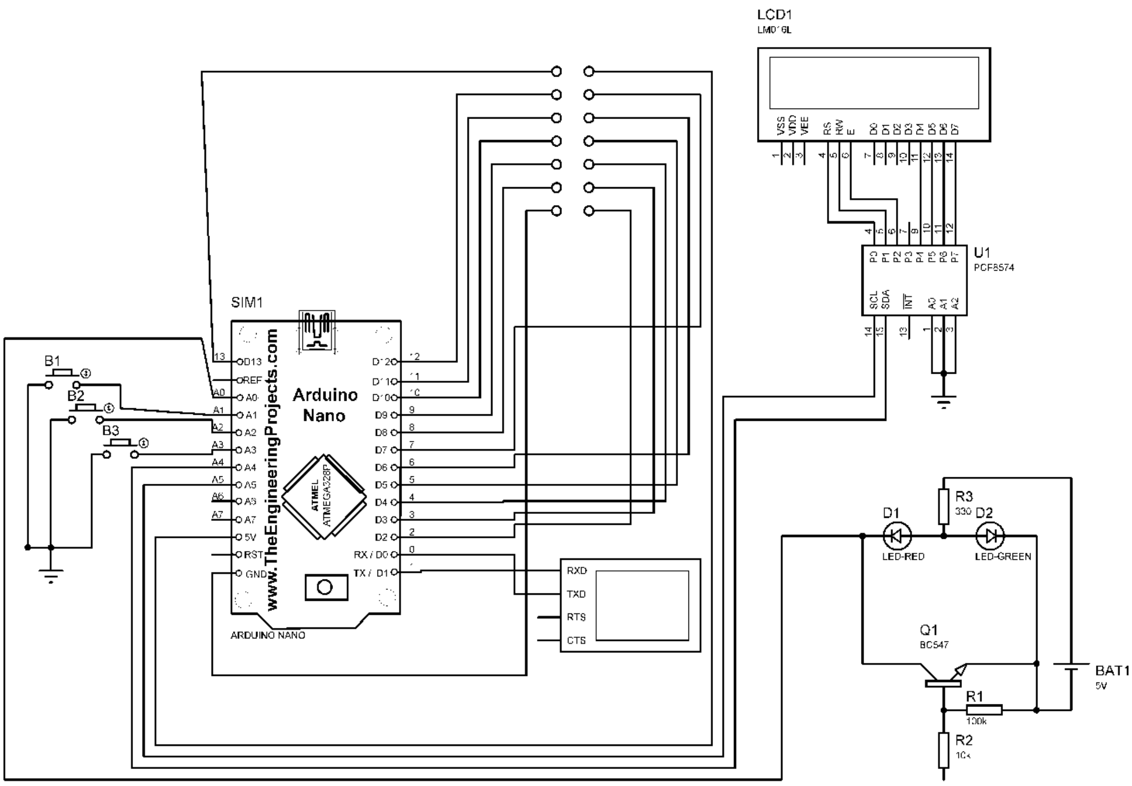


Figure 5.1 Circuit Diagram of Proposed System

The core system components of the Microcontroller Based Digital Logic Integrated Circuit Tester and Short Circuit Detector consist of the following parts:

- Arduino Nano
- 16x2 I2C LCD
- IC Socket
- Push Button
- Power Supply Unit
- BC547 Transistor
- LED
- Resistor

5.2.1 Arduino Nano

Our Component Tester project's central processing unit (MCU) is a small, flexible Arduino Nano microcontroller board. The Atmel ATmega328P microprocessor

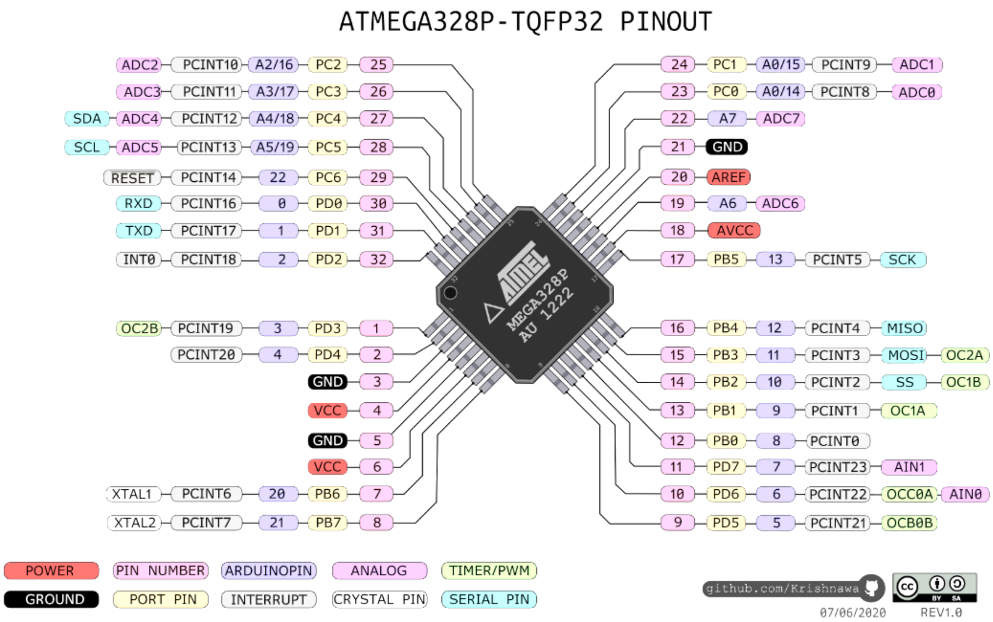


Figure 5.2 Pinout of ATMEGA 328P Microprocessor

forms the basis of the Arduino.cc-developed Nano, which has a compact form factor and is ideal for embedded applications with limited space. Let's take a closer look at its salient characteristics and functionalities:

An Arduino Uno can be utilized as a substitute MCU. A flexible microcontroller board built around the ATmega328P chip is the Arduino Uno. Because of its sturdy construction and ease of use, this Arduino board is among the most popular and beginner-friendly models. The Uno has a 16 MHz quartz crystal, 6 analog inputs, 14 digital input/output pins, a USB port, a

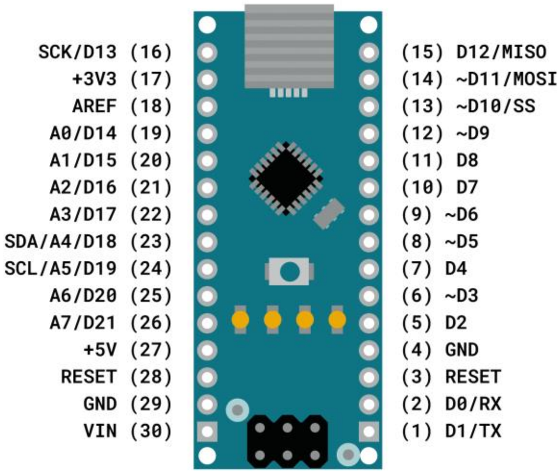


Figure 5.3 Overview of Arduino Nano

power jack, and a reset button. It can be powered by an external power source or a USB connection. The Uno may be used for a variety of projects, from straightforward prototypes to intricate robotics, automation, and Internet of Things applications, because it can be used with a broad selection of sensors, actuators, and shields. It's a great option for novice and seasoned builders alike because of its uniform layout and robust community support. [12]

5.2.2 16x2 LCD

A popular alphanumeric display in embedded systems, such as our Tester, is the 16x2 LCD (Liquid Crystal Display) module. It has a 2-line, 16-character wide display that can show up to 32 characters at once. Every character is composed of a matrix of 5 by 8 pixels.

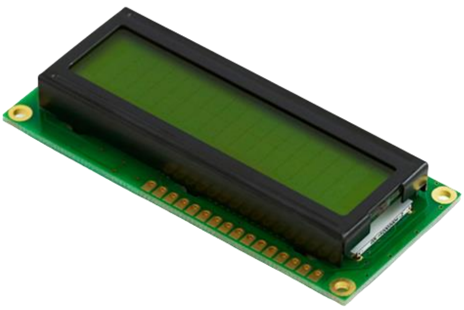


Figure 5.4 16x2 LCD Display

LCD modules typically feature a standard character set, including letters and symbols, with some allowing for custom characters. The microcontroller sends commands as binary data to configure the display, adjusting properties like screen clearing and cursor positioning. The 16x2 LCD module is commonly used to show menu options, test results, and real-time feedback; in our project, it provides clear information for users of the Tester.

5.2.3 I2C Module

Our Tester is not complete without the I2C (Inter-Integrated Circuit) module, which allows serial device communication. I2C uses a master-slave architecture and runs on just two wires: the serial data line (SDA) and the serial clock line (SCL). In this scenario, one device controls communication as the master, and other devices function as slaves, carrying out orders. Because every slave device has an own 7-bit address, the master can choose which devices to communicate with.

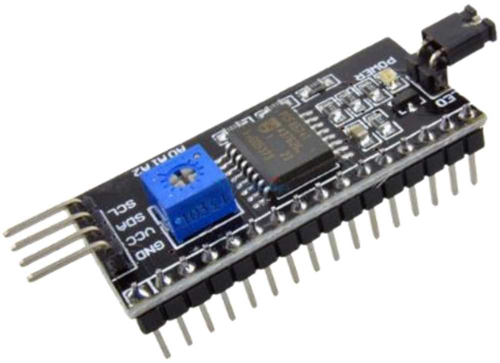


Figure 5.5 I2C Module

In I2C, data is transferred via frames or packets that are started and stopped by the master device. To provide dependable communication, these frames include data bytes, acknowledgment signals, and an address byte. Data integrity and precise timing between devices are guaranteed by clock synchronization, which is controlled by the clock signal of the master. I2C facilitates multi-device communication on a single bus, ideal for space-constrained applications with minimal wiring. In this project, the Arduino Nano and 16x2 LCD unit communicate seamlessly via I2C, enabling efficient data transmission, simplified wiring, and resource conservation.

5.2.4 Power Adapter

The power supply selected for our project is a USB adapter that consistently outputs one ampere of direct current (DC) at five volts. The electrical power required to run the different parts of our Tester is mostly dependent on this adaptor.

The power adapter ensures compatibility with the voltage requirements of most electronic equipment, including microcontroller boards, display units, and other peripherals, by providing a stable output voltage of 5 volts. This standardized voltage level allows for smooth and continuous operation, while the current output of 1 ampere (1A) meets the operational demands of connected components.

Figure 5.6 Samsung 5W Power Adapter

Featuring a USB interface, the power adapter is user-friendly and versatile, allowing connections to standard USB ports on laptops, desktop computers, USB power banks, and wall adapters. This flexibility enhances accessibility and simplifies connectivity. Additionally, the power adapter may include safety features such as short circuit, overvoltage, and overcurrent protection, ensuring safe and reliable operation while safeguarding both the power supply and connected devices.

5.2.5 Buttons

In our project, the push button functions as a basic input unit that allows users to communicate with the Tester. These tactile switches provide simplicity and adaptability by facilitating user control and command execution within the system.

Figure 5.7 Push Button

The ability to select modes in the Tester is a crucial feature of the push buttons. These buttons facilitate test initiation, menu navigation, and option selection. To ensure reliable operation, the Arduino input pin can be configured with internal or external pull-up or pull-down resistors. Using the pinMode function with INPUT_PULLUP, one terminal of the push button connects to ground while the other connects to the digital input pin, eliminating the need for an external resistor. Alternatively, an external pull-down configuration requires a 10k ohm resistor between the digital input pin and the 5V supply for a logic low reading when the button is not pressed. While both methods are effective, internal pull-up resistors are often preferred for their simplicity. The choice depends on your project's specific needs.

5.2.6 DIP Socket

In electronic circuits, a DIP (Dual In-Line Package) socket is a type of socket that makes it easier to insert, remove, and replace integrated circuits (ICs) without having to solder them directly onto the printed circuit board (PCB). In order to replicate the pin arrangement of DIP ICs, DIP sockets are made of a plastic housing with two rows of pins arranged parallel to one another. To fit different IC package sizes, these sockets come in a variety of pin counts, including 8, 14, 16, and 20-pin.

Figure 5.8 16 Pin DIP Socket

A DIP socket's main purpose is to connect an integrated circuit (IC) to a printed circuit board (PCB) securely and reliably while making it simple to remove or replace the IC as needed. This is especially helpful for electronic projects in the prototype, testing, and troubleshooting phases, when it's critical to be able to replace ICs fast without causing PCB damage.

DIP sockets are widely used in a wide range of electronic applications, such as consumer electronics, industrial automation, hobby projects, and prototyping. They work with many different DIP ICs, including operational amplifiers, microcontrollers, and analog-to-digital converters. In general, DIP sockets are essential for making it easier to integrate and maintain integrated circuits (ICs), which improves the effectiveness and dependability of electronic systems.

5.2.7 BC547 Transistor

The BC547 transistor is an NPN transistor. A transistor is nothing but the transfer of resistance which is used for amplifying the current. A small current of the base terminal of this transistor will control the large current of emitter and base terminals. The main function of this transistor is to amplify as well as switching purposes. The maximum gain current of this transistor is 800A. The BC547 transistor includes three pins which include the following.

- Pin1 (Collector): The flow of current will be through the collector terminal.
- Pin2 (Base): This pin controls the transistor biasing.
- Pin3 (Emitter): The current supplies out through emitter terminal

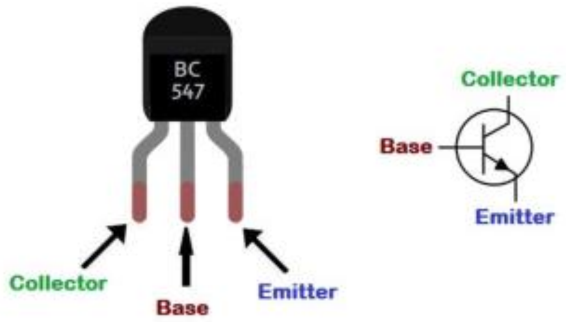


Figure 5.9 Pin Configuration of BC547 Transistor

A Transistor works as an amplifier while functions in the active region to amplify voltage, current, and power at various configurations. The working states of BC547 transistor includes Forward Bias and Reverse Bias. In a forward bias mode, the two terminals like emitter & collector are connected to allow the flow of current through it. Whereas in a reverse bias mode, it doesn’t allow the flow of current through it because it works as an open switch.

5.2.8 Printed Circuit Board (PCB)

A printed circuit board (PCB) is vital in electronic systems, serving as a substrate for mounting and connecting electrical components. It provides a structured framework for creating circuits and ensuring efficient signal routing. PCBs are made by applying conductive copper to a non-conductive substrate, like fiberglass (FR4), with traces connecting the components. Different PCB types are designed for various applications and manufacturing processes.

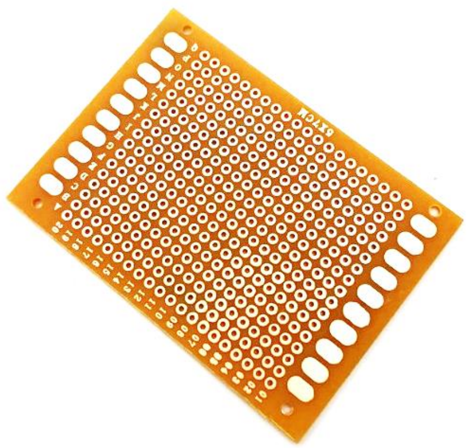


Figure 5.10 Veroboard or Prototype Board

A prototype board, also called a Veroboard, is a fundamental tool used in the electronics prototyping process. These boards have a grid of holes where cables can be used to connect and insert electronic components. Due to their adaptability, prototype boards allow for rapid circuit configuration iteration and debugging.

5.2.9 CAD Software

In developing the Microcontroller-Based Tester, Computer-Aided Design (CAD) software is essential for efficient circuit design and prototyping. I used Proteus for circuit design and simulation, while EasyEDA was employed for creating the PCB layout.

Proteus: This tool, developed by Labcenter Electronics, is popular for its user-friendly interface and extensive component library. It allows for circuit simulation to model behavior before prototyping and supports professional PCB layout transformations, including trace routing and generating manufacturing files like BOM and Gerber. Proteus also enables co-simulation with platforms like Arduino and Raspberry Pi.

EasyEDA: A powerful web-based tool, EasyEDA simplifies PCB design with an intuitive drag-and-drop interface. It features automatic routing and design simulations to ensure functionality before production. Its cloud-based nature facilitates collaboration, allowing users to share designs and work remotely, while also integrating manufacturing services for direct PCB ordering.

5.3 Flow Diagram for Coading

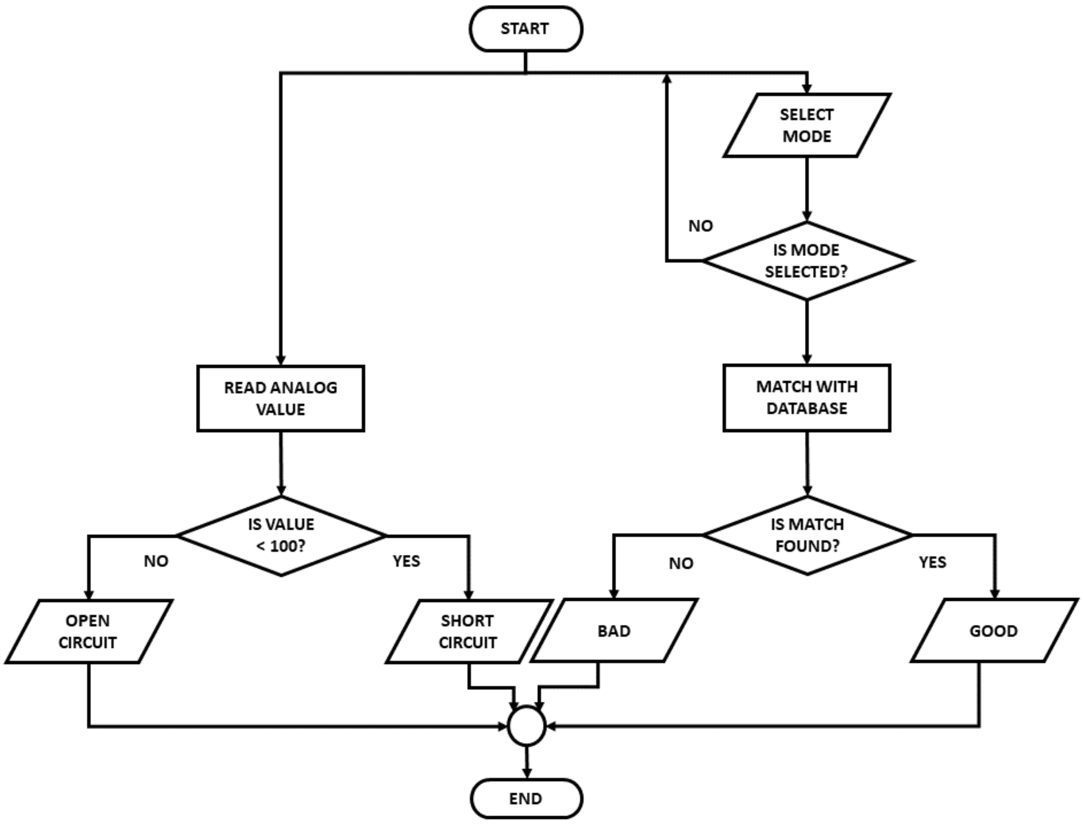


Figure 5.11 Flow Chart to Program the Microcontroller

A flowchart is a graphical representation of a system, method, or process that shows the decisions or actions involved in the process using different shapes and arrows. It acts as a visual aid to help comprehend, evaluate, and explain complicated processes in an easy-to-understand and organized way. A wide range of industries, including computer programming, engineering, corporate management, and education, heavily rely on flowcharts.

The project code is structured around three distinct modes, each designed to test different types of logic gates in digital ICs. The first mode focuses on testing NOT gates. It sets up the necessary inputs, computes the output based on the logic provided by the user, and displays the results on the screen to confirm if the NOT gate is functioning correctly.

The second mode is dedicated to testing NOR gates. Similar to the first mode, it takes user-provided inputs, calculates the corresponding outputs according to the NOR gate truth table, and displays the results. This ensures that NOR gates are working as expected.

In the third mode, the code allows users to select from a variety of logic gates such as AND, OR, XOR, and NAND. Users input the required signals, and the system calculates and displays the output, matching it with the corresponding truth table for each gate type.

The flowchart accompanying the code outlines how each mode operates, allowing users to seamlessly navigate through the testing process. After each test, the system provides feedback on whether the IC is functional or faulty, and users have the option to either restart the testing procedure or exit the program, ensuring ease of use and flexibility in operation.

5.3.1 Coading with Arduino

Arduino can be coded in C/C++ language. The coding screen is divided into two blocks. The setup is considered as the preparation block, while the loop is considered as the execution block. The set of statements in the setup and loop blocks are enclosed with the curly brackets. We can write multiple statements depending on the coding requirements for a particular project. [13]

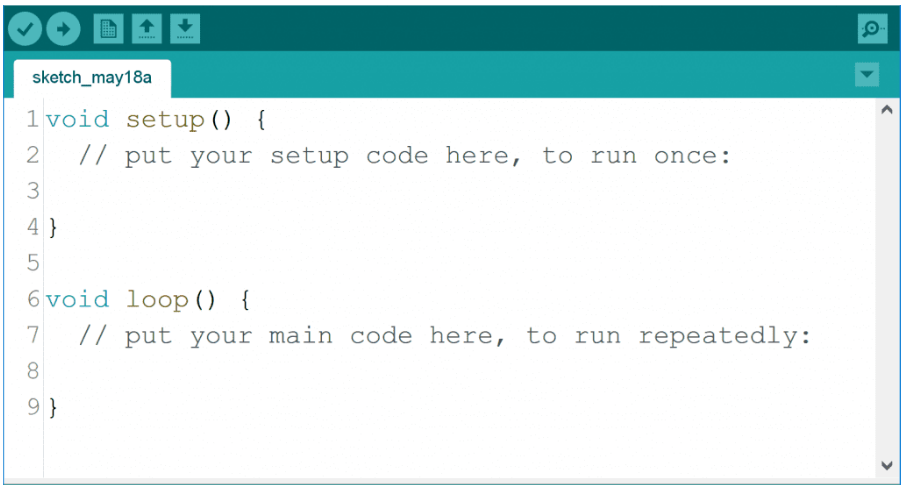


Figure 5.12 Basic Arduino IDE Interface

void setup(): This function initializes the code that runs once when the Arduino powers up or resets. It sets pin modes, libraries, and variables.

void loop(): This function contains code that executes repeatedly, allowing for continuous operation based on variable values.

delay(milliseconds): A blocking function that pauses the program for a specified duration in milliseconds (1 second = 1000 milliseconds). While useful for simple tasks, it can hinder multitasking.

pinMode(pin, mode): Configures a specific pin as either INPUT or OUTPUT. The syntax is pinMode(pin, mode) where pin is the pin number and mode is either INPUT or OUTPUT.

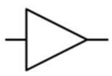
digitalWrite(pin, value): Sets a pin to HIGH or LOW. If the pin isn't set as OUTPUT, devices like LEDs may not function correctly.

digitalRead(pin): Reads the value from a specified digital pin (either HIGH or LOW). The syntax is digitalRead(pin), where pin is the Arduino pin number. Note that analog input pins can also serve as digital pins. [14]

5.4 Methodology

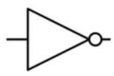
Table 5.1 Truth Table of All Logic Gates

Buffer



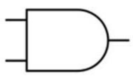
Input	Output
0	0
1	1

Inverter



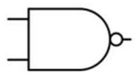
Input	Output
0	1
1	0

AND



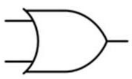
A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

NAND



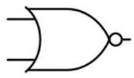
A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

OR



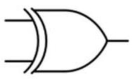
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

NOR



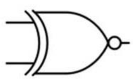
A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

XOR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

XNOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

The " Microcontroller Based Digital Logic Integrated Circuit Tester and Short Circuit Detector" employs a streamlined approach to testing electronic components, emphasizing simplicity, accuracy, and cost-effectiveness. The system is built around the ATMEGA328 microcontroller, which processes input signals and outputs test results. An I2C 16x2 display unit serves as the user interface, providing clear feedback about the component being tested. Additionally, the system includes a socket for secure component insertion and external circuitry with a BC547 transistor for short-circuit detection, enhancing its diagnostic capabilities.

The testing begins when the user inserts a 14-pin IC into the socket and selects one of three modes using push buttons:

- Mode 1: Tests NOT gates.
- Mode 2: Tests NOR gates.
- Mode 3: Tests AND, OR, XOR, and NAND gates.

Once a mode is selected, the microcontroller sends specific binary signals (0s and 1s) to the IC’s gates for functionality testing. It captures the IC’s output and compares it with an embedded truth table stored in memory, which outlines expected outputs for each gate type. If the output matches, the device confirms the IC is functioning correctly, displaying the model and "Good" on the screen. Any discrepancies indicate faulty gates, accompanied by specific error messages.

Additionally, components are tested for short circuits using a BC547 transistor, with red and green LEDs providing visual feedback. The red LED indicates a fault condition, while the green LED confirms proper functionality. This systematic approach ensures accurate testing and quick identification of model numbers and operational status, enhancing the device's utility in engineering labs and academic settings as a reliable solution for testing digital logic ICs.

5.5 Conclusion

In conclusion, the proposed digital logic IC tester presents an effective solution for evaluating the functionality of digital logic integrated circuits. Utilizing an ATMEGA328 microcontroller and an I2C LCD display, the system combines cost-effectiveness with user-friendliness. The structured program flow, featuring distinct modes for testing various logic gates, ensures accurate and efficient assessments.

The tester's capability to compare IC outputs against an embedded truth table guarantees reliable results, making it a valuable tool for educational and practical applications. Additionally, the integration of hardware and software components enhances the overall user experience, providing a dependable solution for testing digital logic ICs. This design not only meets the project's objectives but also serves as a robust resource in engineering labs and academic settings.

CHAPTER 6

IMPLEMENTATION AND RESULT ANALYSIS

6.1 Overview

In this chapter, we will delve into the implementation of the Microcontroller-Based Digital Logic Integrated Circuit Tester and Short Circuit Detector. Building upon the design principles established in previous sections, this implementation phase focuses on integrating the selected components into a functional system.

We will detail the assembly of the circuit, including the ATMEGA328 microcontroller, I2C 16x2 display, and the short circuit detection mechanism using a BC547 transistor. Additionally, we will outline the coding strategies employed to ensure accurate testing and reliable output.

Throughout this chapter, we will present the results of various tests conducted on digital logic ICs, showcasing the device's ability to evaluate different logic gates and detect short circuits effectively. By illustrating the implementation process and demonstrating the outputs, this chapter aims to provide a comprehensive understanding of how the system operates in real-world scenarios, highlighting its utility for engineers, students, and hobbyists in digital electronics.

6.2 Logic IC Tester Implementation

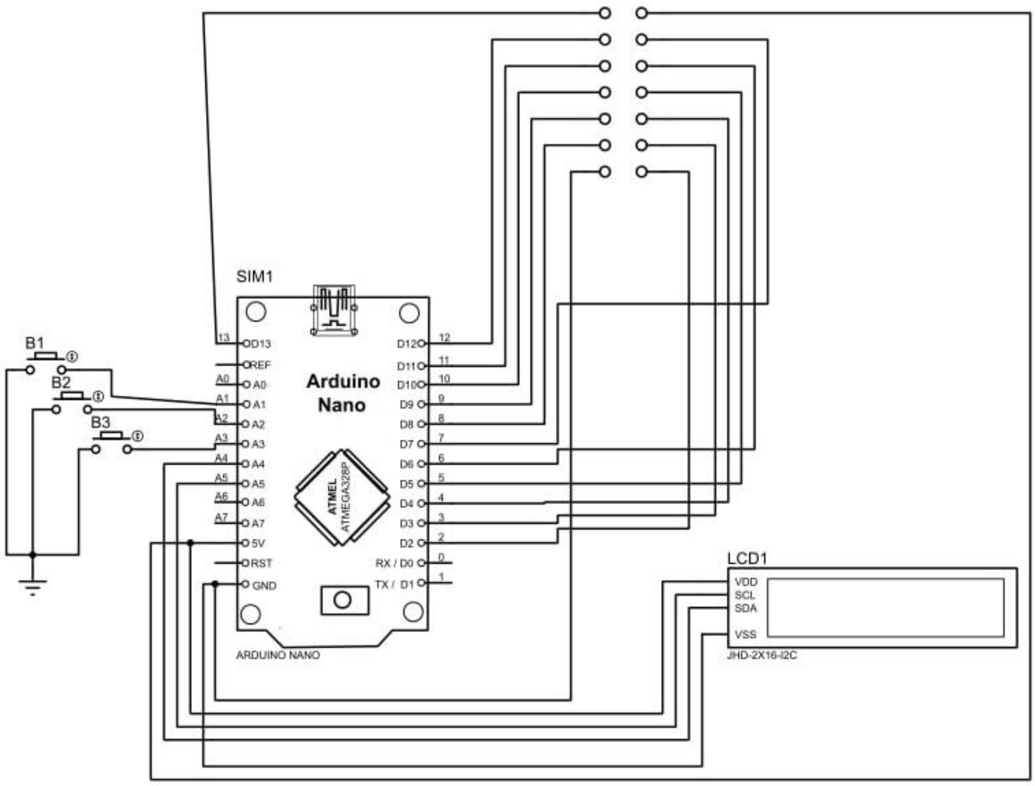


Figure 6.1 Circuit Diagram of Logic IC Tester Part

Table 6.1 Working Principles of Digital Logic IC Tester

Input 4X2		
	C0	C1
R0	0	0
R1	0	1
R2	1	0
R3	1	1

Database of All Gate 4X5					
	C0	C1	C2	C3	C4
R0	0	0	1	0	1
R1	0	1	1	1	0
R2	0	1	1	1	0
R3	1	1	0	0	0

Initialized Output Array 5X4				
	C0	C1	C2	C3
R0	0	0	0	0
R1	0	0	0	0
R2	0	0	0	0
R3	0	0	0	0
R4	5	5	5	5

Output for AND Gate [4][0]				
	C0	C1	C2	C3
R0	0	0	0	0
R1	0	0	0	0
R2	0	0	0	0
R3	1	1	1	1
R4	0	0	0	1

6.3 Short Circuit Detector Implementation

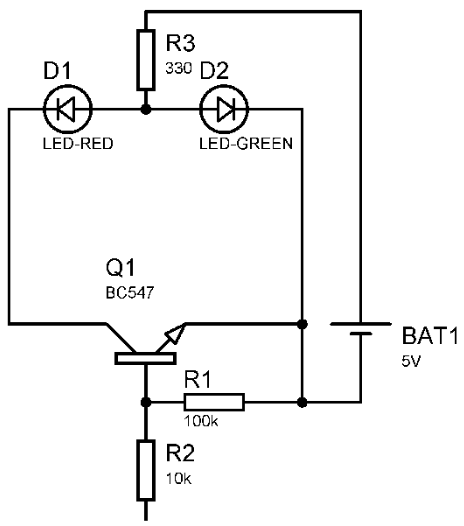


Figure 6.2 Circuit Diagram of Short Circuit Detector Part

Here, the Green LED will glow when the circuit is open and the Red LED will glow when the circuit is short. If we place the two probes in two terminals of a components to test it, the Red LED will glow if there is any short between the terminal.

In case of a diode, it will show short in forward bias condition and open in reverse bias condition. If it shows different output, the diode is faulty.

In case of a capacitor, as there are two plates separated by and insulator, the tester will show open circuit while connecting the two terminals.

6.4 Printed Circuit Board Design

The Printed Circuit Board (PCB) for this project was designed using EasyEDA software, chosen for its user-friendly features in schematic capture and PCB layout.

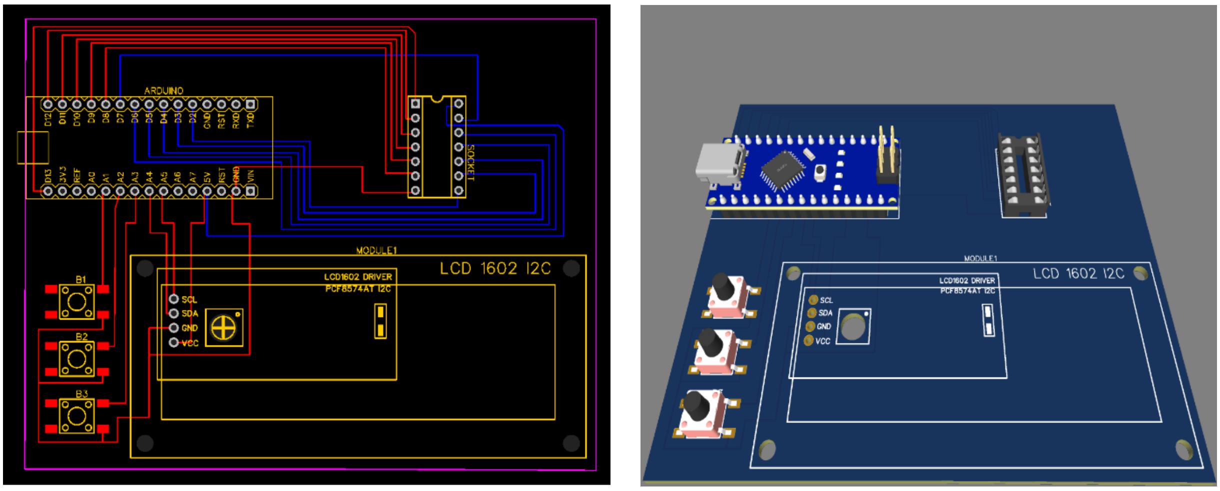


Figure 6.3 Full PCB Layout and 3D Model

The design process included creating the top layer with essential components like the microcontroller and testing sockets, while the bottom layer focused on efficient trace routing to minimize interference. Finally, a 3D model was generated to visualize the fabricated board, ensuring an organized and effective layout.

6.5 Implementation

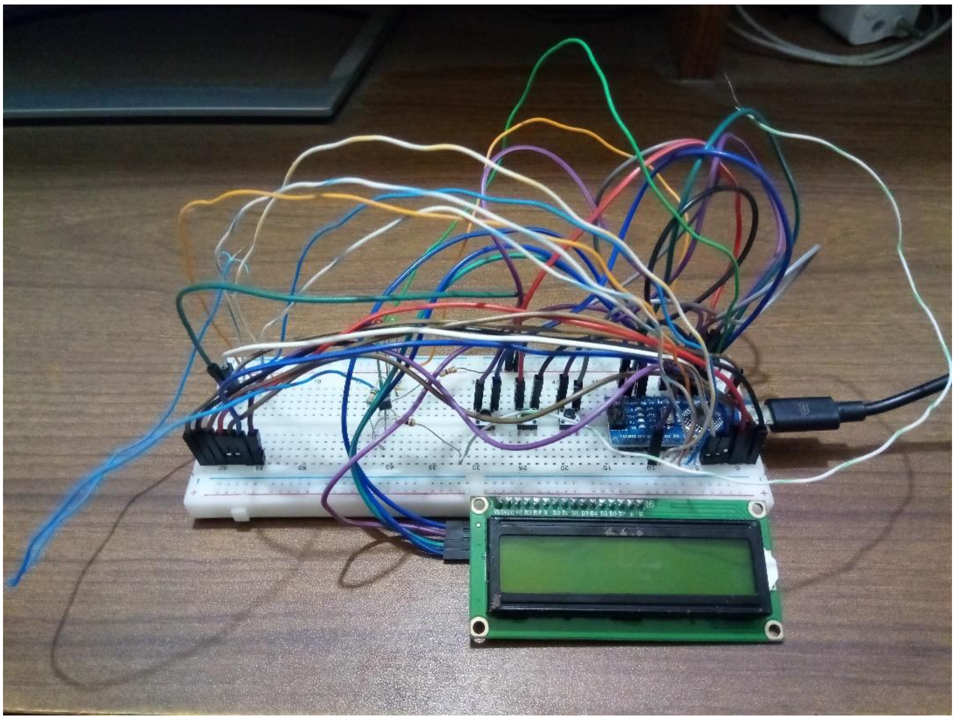


Figure 6.4 Implementation of the Project with Open Circuitry

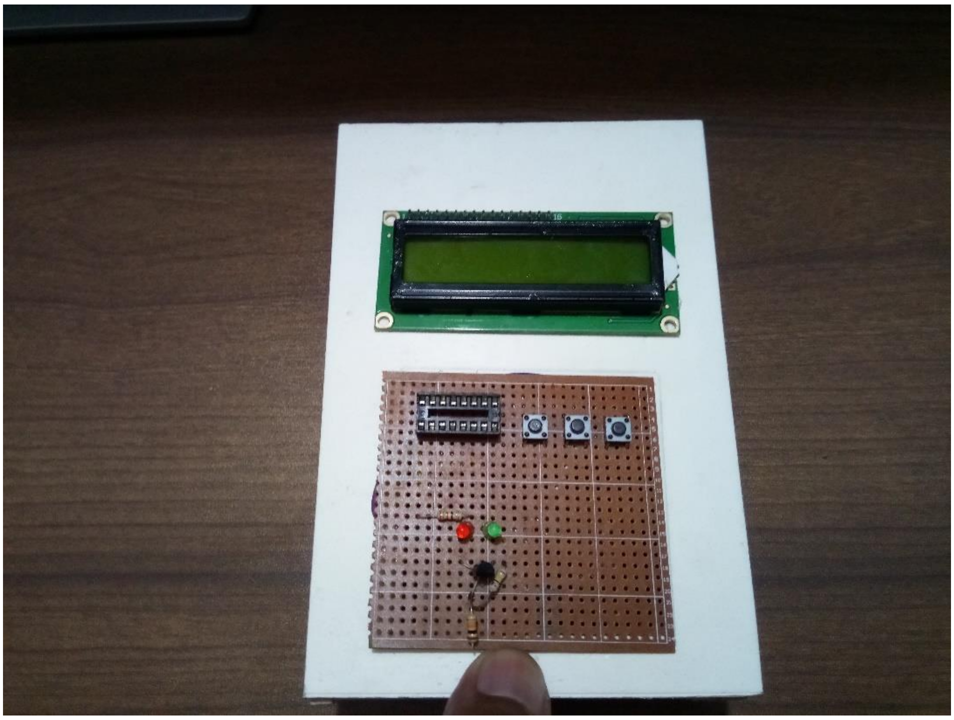


Figure 6.5 Implementation of the Project as a Complete Device

6.5.1 Results and Discussion

The welcome screen will show this info,

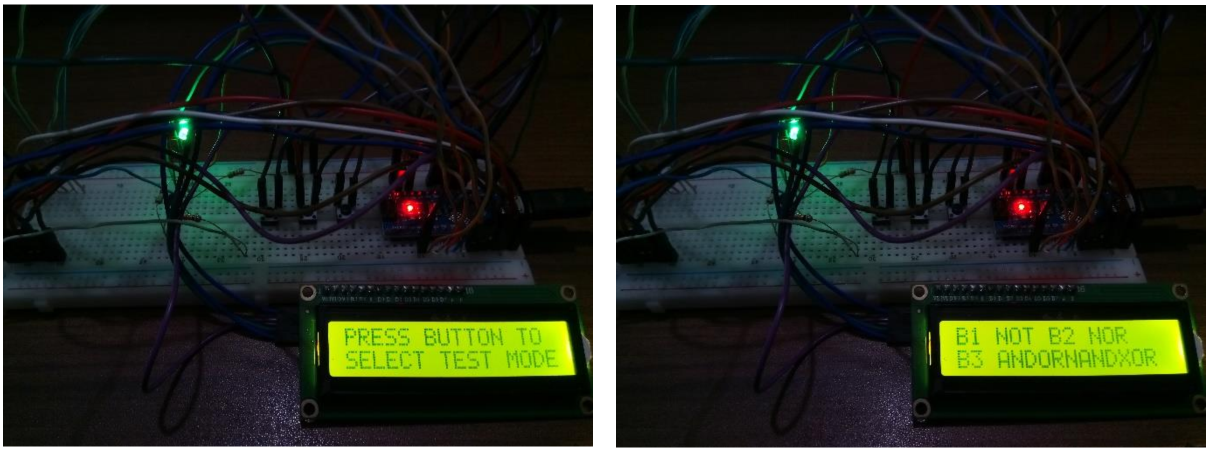


Figure 6.6 Welcome Screen of the Device

The first message will show, “Press Button to Select Mode”. The second message will show, “B1 NOT B2 NOR B3 ANDORNANDXOR”. There are three modes,

- Mode 1: 1st Button from Right, Will check for NOT Gate
- Mode 2: 2nd Button from Right, Will check for NOR Gate
- Mode 3: 3rd Button from Right, Will check for NAND, AND, OR, XOR Gate

Between the two terminals, if there is no connection, the device will show open circuit in LCD. But if there is any connection the LCD will display short circuit.

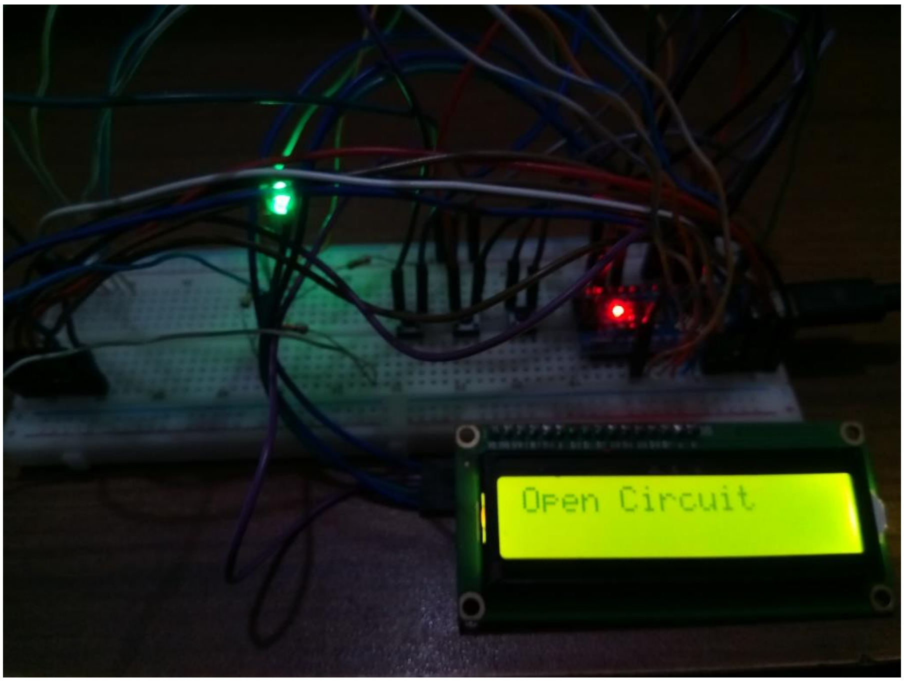


Figure 6.7 Without Connecting the Two Terminals

Also, the Green LED will glow when the circuit is open and the Red LED will glow when the circuit is short. If we place the two probes in two terminals of a components to test it, the Red LED will glow if there is any short between the terminal.

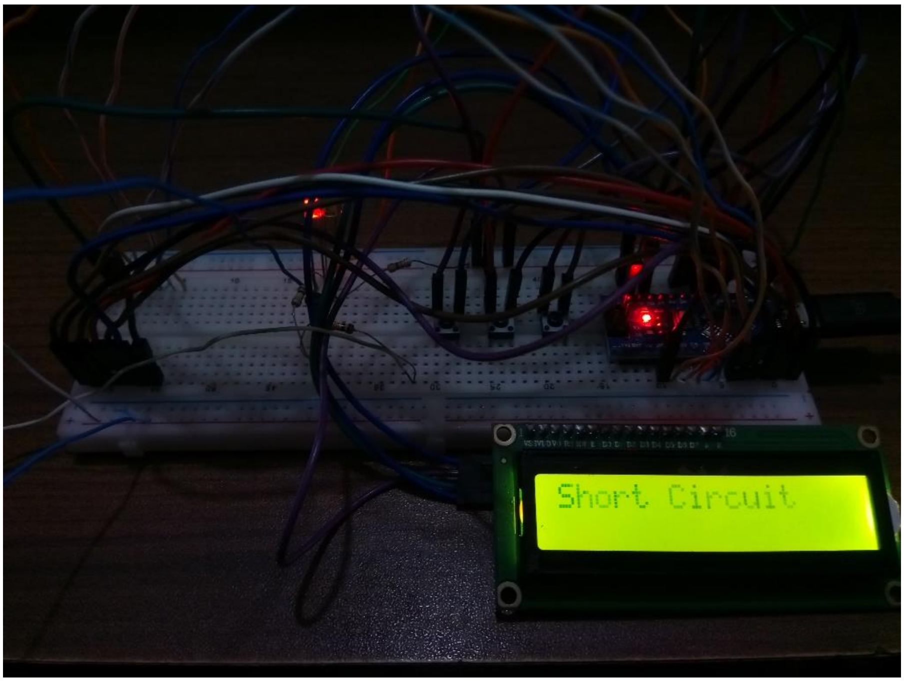


Figure 6.8 Connecting the Two Terminals

Now in case of Digital Integrated Circuit Tester, we’ve connected a faulty NAND Gate IC from our Electronics Lab II and here is the output for Pressing the 2nd Button,

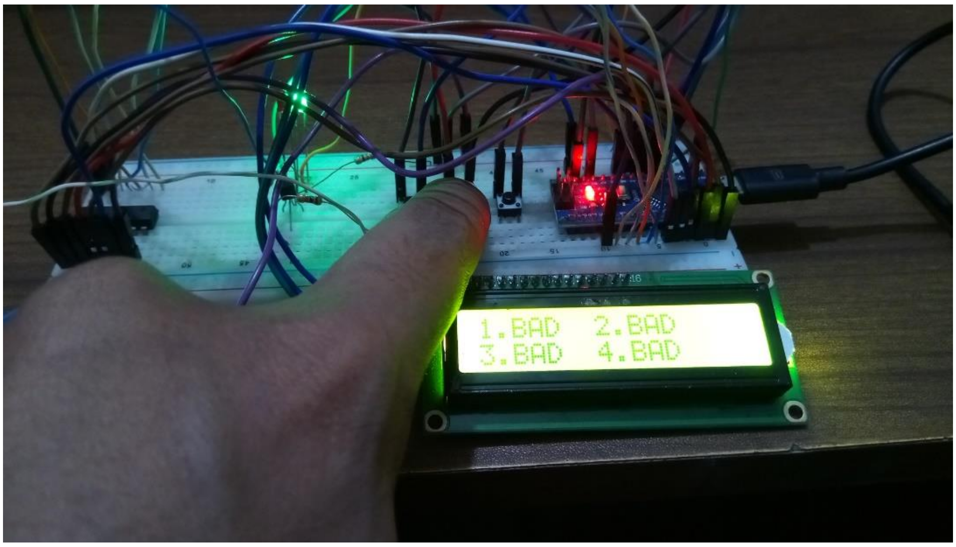


Figure 6.9 Output for Connecting NAND Gate and Selecting Mode 2

The output show, “1. BAD 2.BAD 3.BAD 4.BAD”. That means all the gates are bad as because the Mode 2 is set for only testing the NOR gate ICs. The last message will show, “BAD IC OR NO IC IS CONNECTED”. Now, a faulty IC is inserted,

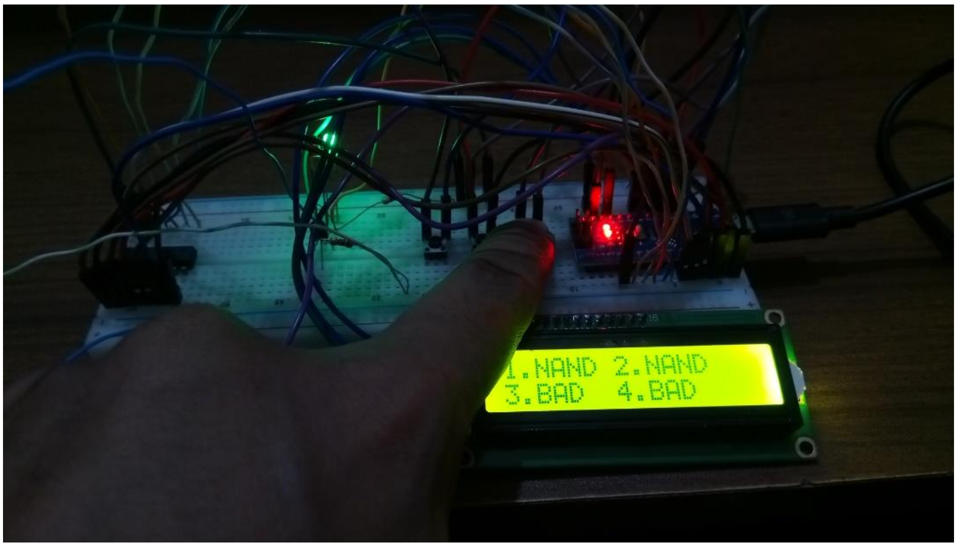


Figure 6.10 Output for Connecting NAND Gate and Selecting Mode 3

As we’ve mentioned before the gate was faulty earlier, so it’s showing two gates are NAND and remaining two of them are BAD. For this insertion the final message will show “ALL GATES ARE NOT FUNCTION”.

Now we’ve connected a new NOR Gate IC bought from the electronics market to ensure the IC is good to get accurate result. Selected Mode 2 for testing the NOR gate IC (7402).

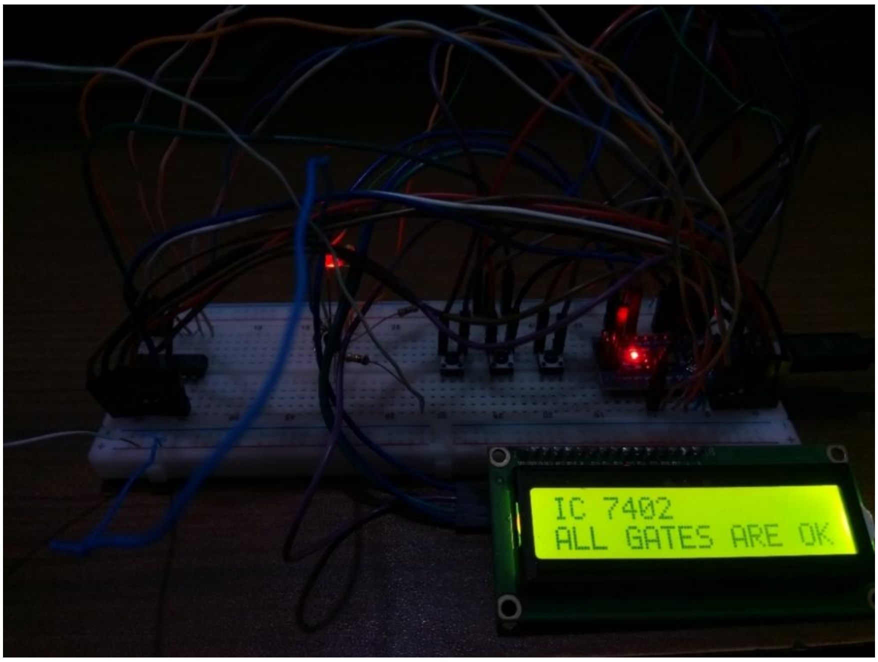


Figure 6.11 Output for Connecting NOR Gate and Selecting Mode 2

As the IC is supposed to be good, the output is showing the name of the all four gates as well as the IC model in the LCD display.

6.6 Cost Analysis

The cost analysis is done by comparing the total cost of the device with the price of the available devices in the market. The costing of the device includes the components price and some relative cost. Man power cost is excluded at this moment.

Table 6.2 Pricing of Components Used in the Project

Components	Price
ATmega328P	TK.500
I2C LCD 1602	TK.250
Power Adapter	TK.150
Prototype PCB	TK.150
ZIF Socket	TK.100
IC, Resistor, Button, Wire, LEDs & Others	TK.200
Total Cost	TK.1350

Table 6.3 Costing of Available Devices in the Market

Available Device in Market	Cost
TES210	TK.2350
TES200	TK.3150
G540	TK.6050
PRG-00027	TK.7350

Table 6.4 Price Comparison

Approximate Less Cost	Less Cost
Minimum TK. 1000	Minimum 42.5%

So, the cost of the project is actually 42.5% less than the device available in the market. Though this device had a lot of limitation but it is suitable typical academic use, where the user doesn't need too many components to test.

6.7 Conclusion

In conclusion, the Microcontroller Based Digital Logic Integrated Circuit Tester offers a reliable and cost-effective approach for testing digital logic integrated circuits, effectively addressing the needs of both educational institutions and electronics enthusiasts. The careful design and coding, combined with the simulation in Proteus, have ensured the system's ability to accurately test individual logic gates within ICs. It provides clear feedback on whether the IC is fully functional, partially faulty, or completely defective, allowing users to quickly identify and troubleshoot issues.

The user-friendly interface, facilitated by the I2C 16x2 LCD display, ensures that the system is easy to operate, making it highly accessible to both beginners and experienced users alike. The successful integration of both simulation and real-world testing showcases the robustness of the design and underscores its potential impact in electronics education and hobbyist applications. This chapter has provided a comprehensive overview of the tester's functionality and effectiveness, setting the stage for further exploration of its performance, usability, and future enhancements that could expand its testing capabilities further.

CHAPTER 7

CONCLUSION AND SCOPE OF FUTURE WORK

7.1 Conclusion

The Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester represents a significant advancement in component testing, providing a versatile, efficient, and affordable solution for both digital logic ICs and various types of transistors. By integrating an ATMEGA328 microcontroller and a 16x2 I2C LCD display, the device performs accurate, real-time assessments of 14-pin logic ICs with gates such as AND, OR, NAND, NOR, XOR, and NOT, and includes a short-circuit detection feature that enables reliable transistor testing. This combination of IC and transistor testing capabilities addresses key limitations of existing tools, making the tester valuable for a range of users, including students, hobbyists, technicians, and engineers.

This project overcomes the high cost, restricted functionality, and bulkiness of traditional testers by providing a compact, user-friendly device that consolidates multiple testing functions into one. The device’s simplicity and ease of use make it accessible to individuals in resource-constrained environments, such as educational institutions, where it can significantly enhance hands-on learning. Additionally, the clear display and LED indicators simplify diagnostics, allowing users to identify faulty components quickly and efficiently.

Overall, the Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester provides a practical, cost-effective solution for essential component testing, enhancing reliability and supporting the needs of users at various levels in electronics. This tool not only reduces the risk of failures due to faulty components but also encourages broader access to reliable testing equipment, promoting a stronger foundation in electronics for future applications.

7.2 Limitations

The Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester has several limitations affecting its functionality. It is specifically designed for digital logic ICs and cannot test all 14-pin ICs, which restricts its applicability to a narrower range of components. Additionally, the tester may experience calibration issues that can impact measurement accuracy, requiring regular calibration for reliable performance.

While the device provides a useful framework for assessing component functionality, the results are not guaranteed to be 100% accurate. Factors such as component variations and environmental conditions can lead to discrepancies in readings, necessitating caution in result interpretation. These limitations highlight the need for further development to enhance the tester's capabilities and accuracy.

7.3 Scope of Future Work

Future expansions could enhance the tester's functionality to include additional component types, such as operational amplifiers, capacitance, and inductance measurements, making it an even more comprehensive tool for electronics testing and education. In conclusion, the Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester offers a practical and impactful solution, fostering accessibility and innovation in the field of electronics.

The Microcontroller-Based Digital Logic Integrated Circuit and Transistor Tester has substantial potential for future enhancements, with planned upgrades to expand its functionality across a broader range of components and applications. A primary enhancement is integrating operational amplifier (Op-Amp) testing, utilizing the inverting amplifier principle. In this setup, an inverted signal would be fed as input, and the microcontroller would use a time-stamped signal check to determine whether the signal is inverted correctly, confirming the Op-Amp's functionality. The input signal could originate from the microcontroller itself or an external generator, providing flexibility in testing different Op-Amp configurations. Another major upgrade involves adding an LCR meter to measure inductance, capacitance, and resistance, enabling precise measurements for a wider range of components beyond just digital ICs. This feature would significantly broaden the tester's applications, making it valuable for analyzing analog and mixed-signal circuits, as well as digital logic components.

User experience will also be enhanced through the addition of an automated keypad interface, which would streamline control, navigation, and menu selections for faster operation. Paired with this, a graphical or advanced LED display is planned, offering clearer, more detailed output that could display icons, component types, and status information intuitively. In the long term, these improvements aim to evolve the project into a comprehensive diagnostic tool suitable for educational, professional, and hobbyist applications. By consolidating a range of testing functions into a single, user-friendly device, this upgraded tester will provide an invaluable, all-in-one solution for electronics testing.

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APPENDIX

Here is a basic outline of the Arduino code for the project, the Microcontroller Based Digital Logic Integrated Circuit and Transistor Tester, focusing on testing digital logic ICs and short circuit detection using the ATMEGA328 microcontroller, BC457 transistor and the I2C 16x2 LCD display. The code includes a menu for selecting the gate type (NOT, NOR, AND, OR, XOR, NAND) using Buttons and testing the IC accordingly.

The Arduino Code:

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
// Define global variables and arrays
int gateInput1, gateInput2, i, j, k;
//To input the logical value
bool inputTable[4][2] = {0, 0, 0, 1, 1, 0, 1, 1};
//Initializing the output array where output logical values will
be stored
int outputTable[5][4] = {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 5, 5, 5, 5};
//Database of all logic gate ICs output
bool foundMatch, database[4][5] = {0, 0, 1, 0, 1, 0, 1, 1, 1, 0,
0, 1, 1, 1, 0, 1, 1, 0, 0, 0};
// To store the IC name & model
String icName, icModel;

void setup() {
  // Initialize the serial communication
  Serial.begin(9600);
  pinMode(A0, INPUT);
  pinMode(A1, INPUT_PULLUP);
  pinMode(A2, INPUT_PULLUP);
  pinMode(A3, INPUT_PULLUP);
  // Clear the LCD screen
  lcd.clear();
  // Initialize the LCD display with 16 columns and 2 rows
  lcd.begin();
  //Set cursor in first column, first row
  lcd.setCursor(0, 0);
  lcd.print("PRESS BUTTON TO");
  //Set cursor in first column, second row
  lcd.setCursor(0, 1);
  lcd.print("SELECT TEST MODE");
  delay (1000);
```

```
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("B1 NOT B2 NOR");
  lcd.setCursor(0, 1);
  lcd.print("B3 ANDORXORNAND");
  delay (1000);
  // Read and compare analog value from LED
  if (analogRead(A0) < 100) {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Short Circuit");
        Serial.println(analogRead(A0));
} else {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Open Circuit");
        Serial.println(analogRead(A0));
} delay (1000);
}

void loop() {
  //MODE 1 NOT B1
  if(digitalRead(A1) == LOW) {
    // Configure various pins as input and output
    pinMode(13, OUTPUT);
    pinMode(12, INPUT);
    pinMode(11, OUTPUT);
    pinMode(10, INPUT);
    pinMode(9, OUTPUT);
    pinMode(8, INPUT);
    pinMode(7, OUTPUT);
    pinMode(6, INPUT);
    pinMode(5, OUTPUT);
    pinMode(4, INPUT);
    pinMode(3, OUTPUT);
    pinMode(2, INPUT);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Testing......");
    delay(250);
    lcd.clear();
```

```
  digitalWrite(3, LOW);
  digitalWrite(5, LOW);
  digitalWrite(7, LOW);
  digitalWrite(9, LOW);
  digitalWrite(11, LOW);
  digitalWrite(13, LOW);

  int count=0;
  if(digitalRead(12) == HIGH) {
    lcd.setCursor(0,0);
    lcd.print("1.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(0,0);
    lcd.print("1.BAD");
    delay(100);
} if(digitalRead(10) == HIGH) {
    lcd.setCursor(5,0);
    lcd.print("2.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(5,0);
    lcd.print("2.BAD");
    delay(100);
} if(digitalRead(8) == HIGH) {
    lcd.setCursor(10,0);
    lcd.print("3.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(10,0);
    lcd.print("3.BAD");
    delay(100);
} if(digitalRead(6) == HIGH) {
    lcd.setCursor(0,1);
    lcd.print("4.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(0,1);
    lcd.print("4.BAD");
```

```
    delay(100);
} if(digitalRead(4) == HIGH) {
    lcd.setCursor(5,1);
    lcd.print("5.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(5,1);
    lcd.print("5.BAD");
    delay(100);
} if(digitalRead(2) == HIGH) {
    lcd.setCursor(10,1);
    lcd.print("6.NOT");
    delay(100);
    count++;
} else {
    lcd.setCursor(10,1);
    lcd.print("6.BAD");
    delay(100);
} delay(1000);

  if(count==6) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("IC 7404");
    lcd.setCursor(0, 1);
    lcd.print("ALL GATES ARE OK");
    delay(1000);
} else {
    lcd.setCursor(0, 0);
    lcd.clear();
    lcd.print("ALL GATES ARE");
    lcd.setCursor(0, 1);
    lcd.print("NOT FUNCTIONAL");
    delay(1000);
 }
}
//MODE 2 NOR B2
else if(digitalRead(A2) == LOW) {
  pinMode(13, INPUT);
  pinMode(12, OUTPUT);
  pinMode(11, OUTPUT);
  pinMode(10, INPUT);
```

```
  pinMode(9, OUTPUT);
  pinMode(8, OUTPUT);
  pinMode(7, INPUT);
  pinMode(6, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(4, INPUT);
  pinMode(3, OUTPUT);
  pinMode(2, OUTPUT);

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Testing......");
  delay(250);
for (i = 0; i < 4; i++) {
  // Testing gate 1
  digitalWrite(11, inputTable[i][0]);
  digitalWrite(12, inputTable[i][1]);
  gateInput1 = digitalRead(13);
  outputTable[i][0] = gateInput1;
  // Testing gate 2
  digitalWrite(9, inputTable[i][0]);
  digitalWrite(8, inputTable[i][1]);
  gateInput2 = digitalRead(10);
  outputTable[i][1] = gateInput2;
  // Testing gate 3
  digitalWrite(6, inputTable[i][0]);
  digitalWrite(5, inputTable[i][1]);
  gateInput1 = digitalRead(7);
  outputTable[i][2] = gateInput1;
  // Testing gate 4
  digitalWrite(3, inputTable[i][0]);
  digitalWrite(2, inputTable[i][1]);
  gateInput2 = digitalRead(4);
  outputTable[i][3] = gateInput2;
}
for (k = 0; k < 4; k++) {
  for (i = 0; i < 5; i++) {
    foundMatch = true;
    // Compare outputTable values with the database
    for (j = 0; j < 4; j++) {
      if (outputTable[j][k] != database[j][i]) {
        foundMatch = false;
        break;
```

```
        }
      } // Store the matching result
        if (foundMatch) {
        outputTable[4][k] = i;
      }
    }
  } for (i = 0; i < 4; i++) {
     // Set the LCD cursor position
     lcd.setCursor((i % 2) * 7, i / 2);
     // Print gate number
     lcd.print(i+1);
     lcd.print('.');
     switch (outputTable[4][i]) {
      case 4:
        icName = "NOR";
        break;
      default:
        icName = "BAD";
        break;
    } lcd.print(icName + "  ");
      delay(100);
 }
  delay(1000);
  // Check if all IC names are the same
  bool allICsSame = true;
  for (i = 1; i < 4; i++) {
    if (outputTable[4][i] != outputTable[4][0]) {
      allICsSame = false;
      break;
    }
 }// Display a message based on whether all IC names are the same
or not
  if (allICsSame) {
    if(icName=="NOR"){
    lcd.setCursor(0, 0);
    lcd.clear();
    lcd.print("IC 7402");
    lcd.setCursor(0,1);
    lcd.print("ALL GATES ARE OK");
    }else if(icName=="BAD"){
    lcd.setCursor(0, 0);
    lcd.clear();
    lcd.print("BAD IC OR NO");
```

```
    lcd.setCursor(0, 1);
    lcd.print("IC IS CONNECTED");
  }
   delay(1000);
 } else {
    lcd.setCursor(0, 0);
    lcd.clear();
    lcd.print("ALL GATES ARE");
    lcd.setCursor(0, 1);
    lcd.print("NOT FUNCTIONAL");
    delay(1000);
    }
 }//MODE 3 AND OR NAND XOR
  else if(digitalRead(A3) == LOW) {
    pinMode(13, OUTPUT);
    pinMode(12, OUTPUT);
    pinMode(11, INPUT);
    pinMode(10, OUTPUT);
    pinMode(9, OUTPUT);
    pinMode(8, INPUT);
    pinMode(7, OUTPUT);
    pinMode(6, OUTPUT);
    pinMode(5, INPUT);
    pinMode(4, OUTPUT);
    pinMode(3, OUTPUT);
    pinMode(2, INPUT);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Testing......");
    delay(250);

  for (i = 0; i < 4; i++) {
    digitalWrite(13, inputTable[i][0]);
    digitalWrite(12, inputTable[i][1]);
    gateInput1 = digitalRead(11);
    outputTable[i][0] = gateInput1;
    digitalWrite(10, inputTable[i][0]);
    digitalWrite(9, inputTable[i][1]);
    gateInput2 = digitalRead(8);
    outputTable[i][1] = gateInput2;
    digitalWrite(7, inputTable[i][0]);
    digitalWrite(6, inputTable[i][1]);
```

```
    gateInput1 = digitalRead(5);
    outputTable[i][2] = gateInput1;
    // Testing gate 4
    digitalWrite(4, inputTable[i][0]);
    digitalWrite(3, inputTable[i][1]);
    gateInput2 = digitalRead(2);
    outputTable[i][3] = gateInput2;
  }
  for (k = 0; k < 4; k++) {
    for (i = 0; i < 5; i++) {
      foundMatch = true;
      for (j = 0; j < 4; j++) {
        if (outputTable[j][k]!= database[j][i]) {
          foundMatch = false;
          break;
        }
      } if (foundMatch) {
        outputTable[4][k] = i;
      }
    }
  } for (i = 0; i < 4; i++) {
     lcd.setCursor((i % 2) * 7, i / 2);
     lcd.print(i+1);
     lcd.print('.');
     switch (outputTable[4][i]) {
      case 0:
        icName = "AND";
        break;
      case 1:
        icName = "OR";
        break;
      case 2:
        icName = "NAND";
        break;
      case 3:
        icName = "XOR";
        break;
      default:
        icName = "BAD";
        break;
  } lcd.print(icName + "  ");
    delay(100);
}
```

```
delay(1000);
bool allICsSame = true;
for (i = 1; i < 4; i++) {
  if (outputTable[4][i] != outputTable[4][0]) {
    allICsSame = false;
    break;
  }
} if (allICsSame) {
  if(icName=="AND") {
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("IC 7408");
  lcd.setCursor(0,1);
  lcd.print("ALL GATES ARE OK");
  } else if(icName=="OR") {
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("IC 7432");
  lcd.setCursor(0,1);
  lcd.print("ALL GATES ARE OK");
  } else if(icName=="NAND") {
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("IC 7400");
  lcd.setCursor(0,1);
  lcd.print("ALL GATES ARE OK");
  } else if(icName=="XOR") {
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("IC 7486");
  lcd.setCursor(0,1);
  lcd.print("ALL GATES ARE OK");
  }else if(icName=="BAD"){
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("BAD IC OR NO");
  lcd.setCursor(0, 1);
  lcd.print("IC IS CONNECTED");
  } delay(1000);
} else {
  lcd.setCursor(0, 0);
  lcd.clear();
  lcd.print("ALL GATES ARE");
```

```
    lcd.setCursor(0, 1);
    lcd.print("NOT FUNCTIONAL");
    delay(1000);
   }
  }//Or Show INFO Again
   else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("PRESS BUTTON TO");
    lcd.setCursor(0, 1);
    lcd.print("SELECT TEST MODE");
    delay (1000);

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("B1 NOT B2 NOR");
    lcd.setCursor(0, 1);
    lcd.print("B3 ANDORNANDXOR");
    delay (1000);
    if (analogRead(A0) < 100) {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Short Circuit");
        Serial.println(analogRead(A0));
   } else {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Open Circuit");
        Serial.println(analogRead(A0));
    } delay (1000);
   }
 }
```